Ch. 3. Chemical kinetics

THE rate at which a chemical reaction occurs is a critical parameter at an industrial and laboratory-level as the faster the speed the sooner one can commercialize or sell a given chemical product. Slow chemical reactions are, in general, less desirable as it takes more time for the reagents to react. On the other hand, too fast chemical reactions can also be less desirable. Think, for example, in an explosion in which gas and a spark quickly generate carbon dioxide with water and a lot of heat in a way that is not easy to control. This chapter deals with the rate of reactions. You will learn how to calculate what we call a rate law which is a mathematical formula that gives the numerical rate value. You will also learn the collision theory that explains the factors that affect the speed of chemical reactions.

3.1 Rate of reaction

Some chemical reactions happen very quickly such as the explosive reaction between cooking gas and oxygen, happening almost instantaneously. Other reactions happen slowly such as the rusting of iron, taking months to occur. Chemical kinetics is a chemistry discipline that studies the speed of chemical reactions and the factors that affect this measure. The overall goal here is to understand the factors that control the rate of reactions with the ultimate goal of maximizing the generation of products. This section covers the principles of reaction rates, defining two different types of rates, and showing how to interconvert rates of reaction.

Two different ways of defining the reaction rate When a chemical reaction proceeds, reactants decompose into products. Therefore, the concentration of reactants decreases with time as reactant molecules become products, and the concentration of products increases with time as products are being formed. This change in concentration with time is what we call *reaction rate*. There are two different ways of defining the rate of a reaction. We can define the *average reaction rate* $\overline{r}$ as the change in concentration with time or as an *instantaneous rate* r when the time interval is very small (we call this derivative):

$$\boxed{\overline{r} = \frac{\Delta c}{\Delta t} \quad \text{or} \quad r = \lim_{\Delta t \to 0} \frac{\Delta c}{\Delta t} = \frac{dc}{dt}} \tag{3.1}$$

Both types of rates are not numerically the same. The average rate is an average change in concentration between two different times. At the same time, the time interval for the average can be large or small. Differently, the instantaneous rate is calculated as the slope in the concentration vs. time plot at a given time–using mathematical terms we call this the derivative. Instantaneous rates represent more accurately the changes in concentration happening in a reaction. For this reason, from now on, whenever we describe a reaction rate we will refer to the instantaneous rate.

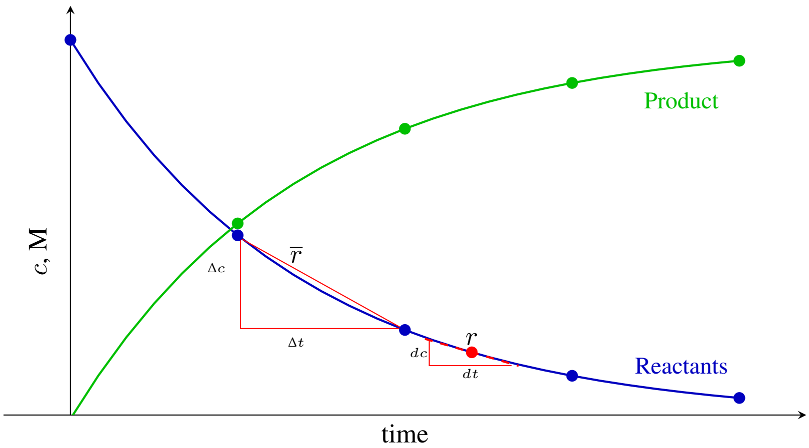


Figure 3.1 Change of the concentration of reactants and products with time showing the average and instantaneous velocity.

Rate of appearance and disappearance At the beginning of a reaction, the molecules of reactants will be consumed while the molecules of products will form (see Figure 3.1). There are three different ways to measure the reaction rate. We can measure the amount of reactants that disappear per unit of time and this would give you the rate of disappearance of a reactant. We can also measure the amount of products formed per unit of time and that would give you the rate of appearance of a product. The numerical signs of a rate are indeed meaningful as rates of disappearance are negative (as the molarity of reactants decreases with time) whereas rates of appearance are positive. Finally, we can measure the amount of reactants that interconvert into products and that would give you the overall rate of reaction, which is always a positive number. Overall, by measuring concentration changes of reactants and products we can compute three types of rates: rates of disappearance, rates of appearance, and rates of reaction. Let us work on a sample reaction:

$$I^-_{(aq)} + OCl^-_{(aq)} \longrightarrow Cl^-_{(aq)} + OI^-_{(aq)}$$

We have that the rate of disappearance of $I^-_{(aq)}$ that should be written as:

$$r_{I^-} = \frac{d[I^-]}{dt}$$

Similarly, we have that the rate of appearance of Cl^- is:

$$r_{Cl^-} = \frac{d[Cl^-]}{dt}$$

As chemical rates represent the change of concentration of a given reactant with time, the units for chemical rate are M/s. Other units involving different units of concentration

Relating rates of appearance and disappearance We can relate the rates of any two reactants or products in a chemical reaction and, at the same time, we can relate the rate of reaction with the rate of any product or reactant–remember there is a single rate of reaction and multiple rates of appearance and disappearance of products and reactants. For example, for the following reaction:

$$4\,NH_{3(g)} + 5\,O_{2(g)} \longrightarrow 4\,NO_{(g)} + 6\,H_2O_{(g)}$$

We can relate r_{NH_3} and r_{H_2O}:

$$-\frac{1}{4}r_{NH_3} = +\frac{1}{6}r_{H_2O}$$

Overall, a negative sign should be included before the rate of any reactant and a positive sign before the rate of any product. You also need to divide by the stoichiometric coefficients of each reactant and product. Similarly, we can relate r_{O_2} with the rate of reaction r:

$$-\frac{1}{5}r_{O_2} = r$$

For a general equation such as:

$$aA + bB \longrightarrow cC + dD$$

we have that the general relationship between any rate and the overall reaction rate is:

$$\boxed{r = -\frac{1}{a}r_A = -\frac{1}{b}r_B = +\frac{1}{c}r_C = +\frac{1}{d}r_D} \tag{3.2}$$

Sample Problem 1

For the reaction below, the rate of disappearance of A is -0.1M/s. Calculate: (a) The rate of disappearance of B. (b) The rate of appearance of C. (c) The rate of appearance of D. (d) The rate of reaction.

$$2\,A + 3\,B \longrightarrow 4\,C + 5\,D$$

SOLUTION

In order to calculate the rate of disappearance of B, we will use the following relationship, in which the rate of disappearance of A and B have a negative sign and carries the respective stoichiometric coefficient

$$-\frac{1}{2}r_A = -\frac{1}{3}r_B$$

We know r_A and hence we can solve for r_B:

$$-\frac{1}{2}(-0.1) = -\frac{1}{3}r_B$$

We have that r_B is -0.15M/s. We will now use a similar relationship to calculate r_C:

$$-\frac{1}{2}(-0.1) = +\frac{1}{4}r_C$$

We have that r_C is +0.2M/s. Finally, we will calculate r_D and r:

$$-\frac{1}{2}(-0.1) = +\frac{1}{5}r_D \text{ and } -\frac{1}{2}(-0.1) = r$$

We have that r_D is +0.25M/s and r is 0.05M/s. Mind the rate of disappearance of reactants are negative, whereas the rate of appearance of products and the overall rate of reaction are positive numbers.

STUDY CHECK

For the reaction below, the rate of appearance of C is 0.3M/s. Calculate: (a) The rate of disappearance of A and B. (b) The rate of appearance of D. (c) The rate of reaction.

$$5\,A + 2\,B \longrightarrow C + 2\,D$$

▶Answer: r_A=-1.5M/s, r_B=-0.6M/s, r_D=0.6M/s, and r=0.3M/s

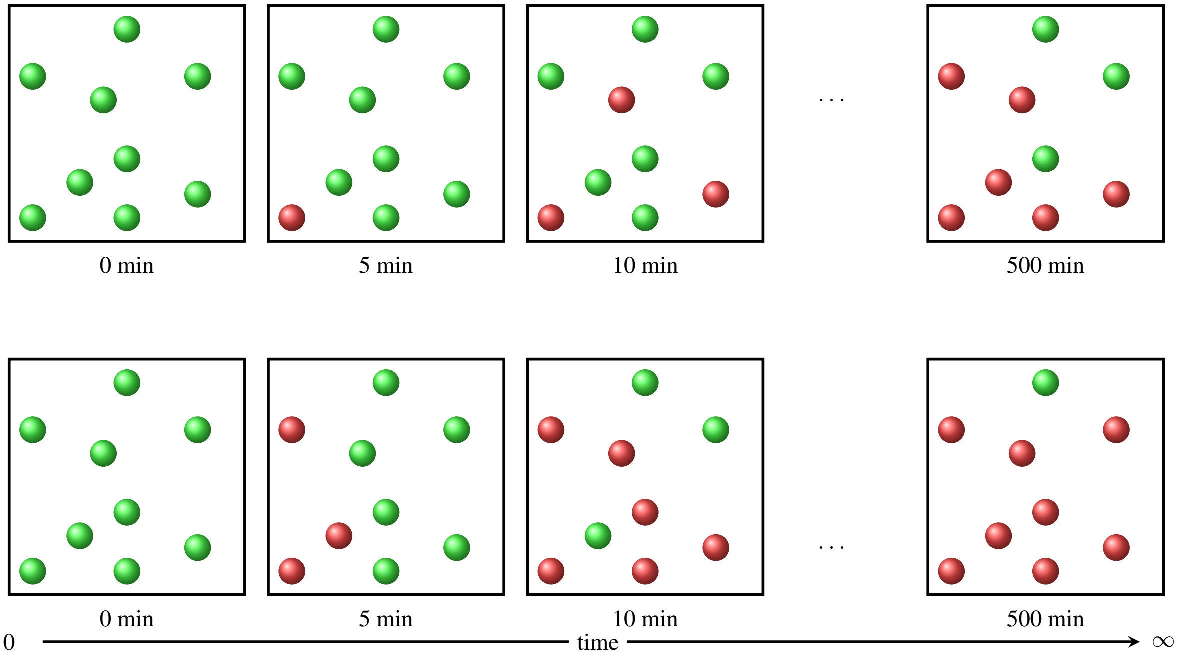


Figure 3.2 A fast (bottom panel) and a slow (top panel) reaction

3.2 Experimental measurement of reaction rates

Reaction rates represent a change in concentration with time. To measure these rates in a laboratory we need to track the changes in concentration happening with time. Spectrophotometry methods help us track such changes in the laboratory. As reaction rates change as well with time, it is convenient to measure rates at the very early stages of the reaction, these are the so-called initial rates, when those values are maximal. Statistical tools or simplified numerical methods to compute slopes can help us contain numerical values of reaction rates. This section addresses the measurement of velocities in the laboratory.

Concentration vs. time plots The concentration of reactants decrease with time, whereas that for products increase. In order to measure reaction rates in a laboratory we would need to pick up either a reactant or a product in order to track concentration changes with time. Here we will focus on reactants and we will explore how the numerical changes on concentrations can be translates into reaction rates. Figure 3.3 display the changes in concentration for a general reactant. We can see how the changes are more pronounced at early times and slowly become flat a late times. This is because any reaction tends to slow down as time progresses. The instantaneous rate of reaction r is just the slope on the graph at a given time.

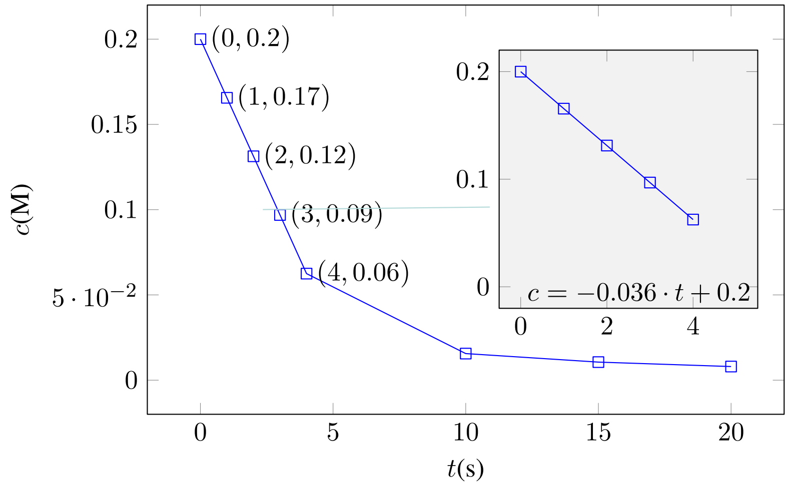


Figure 3.3 Concentration vs. time graph used to calculate initial velocities.

`Initial velocities` As r changes with time, it is convenient to measure this value at very early times, when the concentration changes linearly with time. This is called the *initial rate* of a reaction, r_0, the rate measured very early in the reaction progress. At the same time, *parasitic chemical reactions*, unwanted chemical reactions that occur alongside the primary, desired reaction, do not exist at early times, hence helping to make sure our measured concentration changes are due only to the reaction under study. To obtain initial rates, we will first visually identify the portion of the graph that appears linear. In Figure 3.3, the linear plot exists between the (0,0.2) and (4,0.06) points. There are two different ways to calculate this slope. In a first, less accurate, method we compute the difference in concentration Δc (-0.16) and the difference in time Δt (5) in the linear region. By computing the ratio between Δc and Δt, we will obtain the average initial rate, an estimate of r_0. In Figure 3.3 the average initial rate is

$$\overline{r_0} = \frac{0.06 - 0.20}{4 - 0} = -0.035\text{M/s}$$

A second, more accurate, method uses a linear regression, a statistical function, to fit a line to a linear subset of data points. The slope of the regression is the initial rate. In linear regressions, R-squared represented as r^2 (also known as the coefficient of determination) is a statistic that measures how well a regression model explains the data. In Figure 3.3 the linear regression equation is $c = -0.036 \cdot t + 0.2$ and the initial rate is

$$r_0 = \frac{\mathrm{d}c}{\mathrm{d}t} = -0.032\text{M/s}$$

Sample Problem 2

Answer the following questions using the experimental data given in the table below (a) Graph the data and identify the linear data set (b) Does the data correspond to a reactant or a product (c) Calculate the average initial velocity $\overline{r_0}$ (d) Calculate the instantaneous initial velocity r_0

t (s)	1	2	3	4	6	7	9	10
c (M)	0.40	0.33	0.23	0.14	0.02	0.01	0.005	0.004

SOLUTION

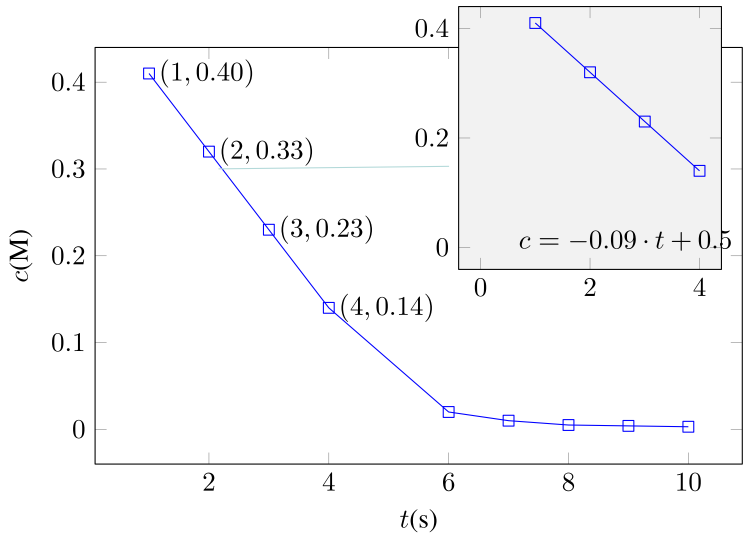
By inspecting the data, we see that the concentration data decreases as time progresses. Hence, we are dealing with a reactant. Also, by inspecting the data, we can see that only the four first data points linearly change with time. For molarities later than 4 seconds, concentration reaches a plateau. In order to calculate initial velocities, we will use only the first data sets. We can calculate the average initial velocity by doing

$$\overline{r_0} = \frac{0.14 - 0.40}{4 - 1} = -0.086\text{M/s}$$

We can calculate the instantaneous initial velocity by interpolating the data with a line. You can do this, for example, in www.graphpad.com/quickcalcs/linear1. The equation of the line is:

$$c = -0.088 \cdot t + 0.5$$

Hence, the instantaneous initial velocity is -0.088M/s.



STUDY CHECK

Calculate the average initial velocity and instantaneous initial velocity from the experimental data given in the table below.

t (s)	1	2	3	4	6	7	9	10
c (M)	0.17	0.24	0.31	0.38	0.50	0.51	0.52	0.53

3.3 Rate law

Some chemical reactions double their rate by increasing the concentration of reactants, whereas, for the same change of concentration, others quadruple their rate. The rate law of a reaction is an experimental formula that explicitly includes the power dependence of the rate of reaction on the concentration of reactants and products. You can find rate laws in a differential form and in an integrated form. On one hand, the differential rate law, or simply the rate law, is convenient when we want to assess the impact of concentration on the rate of reaction. On the other hand, integral rate laws give the time dependency of the concentration, being convenient to track how the reactants disappear with time. This section will cover how to interpret rate laws and how to differentiate rate laws from integral rate laws. It will also introduce the concept of the partial and the total order of reaction, as well as the idea of reaction rate constant.

The rate law The rate law is an experimental expression, a formula that comes from an experiment, relating the rate of reaction with the concentration of the chemicals that impact the rate. An example of a rate law would be:

$$r = k[\mathrm{NO}]^2[\mathrm{O_2}]$$

where:

r is the overall rate of reaction

k is the rate constant

$[\mathrm{NO}]$ is the molarity of NO

$[\mathrm{O_2}]$ is the molarity of O_2

So, what is the purpose of a rate law? Let us use the following rate law as an example:

$$r = 0.002[\mathrm{NO}]^2[\mathrm{O_2}]$$

By means of the rate law above, we can compute the numerical value of the rate of reaction for different concentrations. As such, rate laws help predict the value of the reaction rate.

Partial and overall reaction orders The powers in the rate law are called reaction order. For example, in the rate law below

$$r = k[\mathrm{NO}]^2[\mathrm{O_2}]$$

the order of NO is 2 as the power of NO in the rate law is 2. Similarly, the partial order of O_2 is 1 as the power of O_2 in the rate law is 1. The overall reaction order it 3, and this number results in adding all partial orders. Regarding the way we describe the reaction order, if the reaction order is one, we would say that the reaction is first order. Similarly, if the reaction order is two, we would say the reaction is second order, and so on. The reaction order can also be zero for a zeroth order reaction when the power in the rate law is zero. Reaction orders tend to be integer numbers such as 0, 1, or 2, mostly in educational problems. However, using realistic kinetic data, the partial and total orders can be any number and even have a negative sign. The reaction orders are very important as they indicate how the reaction rate responds to changes in concentration. For example, using the rate law above, if we double the concentration of O_2 the rate law will double, but if we double the concentration of NO, the reaction rate would quadruple, that is, it would be four times larger.

The rate constant, k The rate constant is a critical parameter of a rate law as it tells how the reaction responds to changes on molarity. For example, look this two different rate laws:

$$r_A = 0.01[\mathrm{N_2O_5}] \qquad \text{and} \qquad r_B = 0.001[\mathrm{N_2O_5}]$$

We have that as the rate constant in r_A is larger the changes in the concentration of N_2O_5 will have a stronger impact on the rate than in r_B.

Sample Problem 3

Given the following rate law. Indicate: (a) The rate constant giving the correct units. (b) The partial order of all species. (c) The total order of the reaction. (d) If $[F_2]$ is doubled how would the reaction rate be affected?

$$r = 0.002[Cl_2][F_2]^3$$

Solution

Given the information provided in the rate law, we have that the reaction is first order with respect to Cl_2 and third order with respect to F_2. The overall reaction order is four. The rate constant is 0.002 1/M^{2}s.

Study Check

Given the following rate law. Calculate: (a) The rate constant giving the correct units. (b) The partial order of all species. (c) The total order of the reaction. (d) If $[A]$ is doubled how would the reaction rate be affected?

$$r = 0.056[A]^4[B]^2[C]^2$$

▶Answer: Orders: A(4), B(2), C(2), overall(8); k=0.056 1/M^{7}s; 16 times larger

`The rate constant units` The units of a rate constant are determined by the reaction order. In other words, given a reaction order, we will have a given unit for the rate constant. At the same time, given the units of a rate constant, we can also determine the order of the reaction. The following formula gives the constant units for a given reaction order n:

$$\boxed{\frac{M^{1-n}}{s}} \tag{3.3}$$

where:

n is the reaction order

For example, for first-order reaction (n=1) the units of the reaction constant are 1/s and for a second-order reaction, the units are 1/Ms. For a reaction of zero-order, the units of the constant are M/s. This formula results from a dimensional analysis given that the units of reaction rate r are M/s and the units of concentration is molarity, M. As a side note, here we will use second as a unit of time, however, one can envision a similar formula for different time units.

Sample Problem 4

Given the following rate constants indicate the reaction order: (a) k=0.04 M/s (b) k=0.09 1/M·s (c) k=0.03 1/M^4·s

Solution

We will use Equation 3.3 to calculate the reaction order. We have that 0.04 M/s is order zero, 0.09 1/Ms is second order and 0.03 1/M^{4}s is fifth order.

Study Check

Given the following rate constants indicate the reaction order: (a) k=0.03 1/s (b) k=3 $\times$ 10^{-4} L/s·mol (c) k=3 $\times$ 10^{-5} 1/s·M^2

▶Answer: (a) Order 1 (b) Order 2 (c) Order 3

`Integral reaction law` The reaction law is more specifically called a differential reaction law as it represents a differential equation–this is a mathematical term to describe equations in terms of derivatives. For a differential rate law, there is an integral rate law. The integral rate law is obtained using a mathematical technique called integration–this technique is out of the scope of this class. For example, below we have an example of a differential and integral rate law for a first-order reaction.

$$r = 0.01[\mathrm{N_2O_5}] \qquad \text{and} \qquad \mathrm{Ln}[\mathrm{N_2O_5}] = -0.01 \cdot t + \mathrm{Ln}(0.1)$$

In the differential rate law, we have the explicit reaction rate (r) accompanied by molarities ([N_2O_5]), whereas in the corresponding integral rate law we have molarities ([N_2O_5]) accompanied with time (t). The integral rate law gives the change of the concentration with time, whereas the differential rate law gives the change of the reaction rate with concentration. Each type of law has its purpose and differential rate laws are more common and used when we have experimental data in terms of reaction rates and molarities. The integrated rate law is used when we have data in terms of molarity and time. Overall, differential rate laws and integral rate laws are exchangeable. To convert a differential rate law into its corresponding integral rate law we just need the initial concentration of the species involved in the law, $[\mathrm{A}]_0$. For example, the corresponding differential and integral rate law for a zeroth-order reaction are: Table 3.1 reports the corresponding differential and integral rate laws for different orders.

Table 3.1 Corresponding differential, integral rate laws and half-life formulas

Order	Differential rate law	Integral rate law	Hal-life ($t_{1/2}$) formula
0	$r = k$	$[\mathrm{A}] = -k \cdot t + [\mathrm{A}]_0$	$\frac{[\mathrm{A}]_0}{2k}$
1	$r = k[A]$	$\mathrm{Ln}[\mathrm{A}] = -k \cdot t + \mathrm{Ln}[\mathrm{A}]_0$	$\frac{0.693}{k}$
2	$r = k[A]^2$	$\frac{1}{[\mathrm{A}]} = k \cdot t + \frac{1}{[\mathrm{A}]_0}$	$\frac{1}{k[\mathrm{A}]_0}$

Sample Problem 5

Identify the following rate law as differential or integral and switch into the other form.

$$\frac{1}{[\mathrm{C_2H_6}]} = 4 \times 10^{-4} \cdot t + \frac{1}{0.03}$$

Solution

As the equation contains time this would be a integrated rate law. We can use Table 3.1 to identify the order and compute the differential law. This is a second order reaction with rate constant 4×10^{-4} 1/s, therefore:

$$r = 4 \times 10^{-4}[\mathrm{C_2H_6}]$$

Study Check

Identify the following rate law as differential or integral and switch into the other form for an initial concentration of 0.3M.

$$r = 5 \times 10^{-2}$$

Answer: $[\mathrm{A}] = -5 \times 10^{-2} \cdot t + 0.3$

`Half-life,` $t_{1/2}$ When a reaction starts to happen, the concentration of reactants decreases with time. Half-life is the time at which the initial concentration of reactants is half its initial value. Reactions with small half-life are fast, as they take a short time to reduce the concentration of reactants to half. Differently, reactions with large half-life are slow as they need longer times to proceed. This way, the numerical value of half-life helps you compare the speed of reactions. At the same time, the half-life is a transformation of the rate constant into a time value. That is, by rate constants can be converted into half-lives. There is a single half-life formula for each reaction order and Table 3.1 reports some half-life formulas. If you wonder, where these formulas come from, one can obtain all these formulas by using $[A] = [A]_0/2$ in any integral rate law. For example, for a first order reaction we have that:

$$\mathrm{Ln}[\mathrm{A}] = -k \cdot t + \mathrm{Ln}[\mathrm{A}]_0$$

When $[A] = [A]_0/2$ we have that $t = t_{1/2}$ and therefore:

$$\mathrm{Ln}\left(\frac{[A]_0/2}{[\mathrm{A}]_0}\right) = -k \cdot t_{1/2}$$

that leads to

$$t_{1/2} = -\frac{\mathrm{Ln}(0.5)}{k}$$

Sample Problem 6

Calculate the half-life for a second order reaction with initial concentration of 5.99×10^3 M if the rate constant is 0.005 1/Ms?

SOLUTION

As we deal with a second order reaction, we will use $\frac{1}{k[\mathrm{A}]_0}$ given that $[\mathrm{A}]_0 = 5.99 \times 10^3$M and k=0.005 1/Ms. We have that

$$t_{1/2} = \frac{1}{0.005 \cdot 5.99 \times 10^3} = 0.03s$$

STUDY CHECK

Calculate the half-life for a first order reaction with initial concentration of 5.99×10^3 mol/L if the rate constant is 0.02 1/s?

Answer: 34.65s

3.4 The differential and integral methods for obtaining rate laws

We have addressed the importance of rate laws. However, we have not addressed how to obtain these laws from experiments. This section will cover how to experimentally obtain rate laws by analyzing experimental data. There are two main methods for obtaining rate laws: the differential method and the integral method. Each method differs not only in the nature of the experimental data gathered but also in how this data is processed.

`The differential method` The differential method also called the method of the initial rates measures velocities for different concentrations, and in particular, measures initial velocities. These are velocities at very short times after the reaction starts. This method is convenient when there are parasite reactions that compete with the reaction one wants to study. By using initial rates, we can minimize the impact of competing reactions on the kinetic data. In this method, by comparing different velocities for different concentrations one can calculate the reaction order and the rate constant, that is, one can construct the rate law. The next example will show you how to use this method. As a note, one can only use the differential method when reaction rates data is supplied.

Sample Problem 7

Use the data below to calculate the rate law of the following reaction:

$$A \longrightarrow B$$

Experiment	r (M/s)	$[A]$, (M)
1	0.5	0.5
2	0.72	0.6

SOLUTION

As values of $[A]$ are given, the rate law should be $r = k[A]^n$. In order to calculate the rate law we need the values of k and n. We will apply the rate law two the two different experiments and divide:

$$\left.\begin{array}{ll} 0.5 = k0.5^n & \text{(Experiment 1)} \\ 0.72 = k0.6^n & \text{(Experiment 2)} \end{array}\right\} \frac{0.5}{0.72} = \frac{\cancel{k}}{\cancel{k}} \cdot \left(\frac{0.5}{0.6}\right)^n$$

This approach will cancel k leaving only n.

$$0.694 = (0.833)^n$$

Remember that if we divide two numbers with the same exponent we can combine the numbers and share the exponents. In order to solve for n we need to apply natural logarithms (or simply log) to both sides of the equation in order to move the exponent down:

$$Ln(0.694) = n \cdot Ln(0.833) \text{that gives } n = \frac{Ln(0.694)}{Ln(0.833)} = 1.99 = 2$$

Remember the orders are normally integer numbers so you can round to the closes integer. Now that we have the order, we can use the rate law again to obtain the rate constant using for example: r=0.5M/s and $[A]$ =0.5M, and given that now we know n=2

$$0.5 = k0.5^2$$

Solving for k we have k=2 1/Ms. We have that the reaction law is: $r = 2[A]^2$

STUDY CHECK

Use the data below to calculate the rate law of the following reaction: A $\longrightarrow$ B

Experiment	r (M/s)	$[A]$, (M)
1	0.5	2.5
2	0.6	3.0

Answer: $r = 0.2[A]^1$

The differential method with two reactants This previous example was useful to obtain the rate law using data involving only the concentration of one reactant. In other words, the previous example was useful to obtain simple rate law such as:

$$r = k[A]^x$$

The next example will cover how to calculate the rate law when the data provided involved the concentration of two reactants and therefore the rate law contains the orders of two species. In other words, the next example will show how to obtain more complex rate laws such as:

$$r = k[A]^x[B]^y$$

in which we do not know the reaction constant and two of the order.

Sample Problem 8

Use the data below to calculate the rate law of the following reaction: A + B $\longrightarrow$ C

Experiment	$[A]$ (M)	$[B]$ (M)	r (M/s)
1	0.5	0.2	0.25
2	0.6	0.2	0.36
3	0.5	0.3	0.375
4	0.6	0.3	0.54

Solution

As values of $[A]$ and $[B]$ are provided, the rate law should be $r = k[A]^n[A]^m$. In order to calculate the rate law we need the values of k, n and m. We will apply the rate law two the two different experiments and divide. But this time we will sort the data appropriately. In order to calculate the order or A we will select the data with different values of $[A]$ and the same value of $[B]$:

Experiment	$[A]$ (M)	$[B]$ (M)	r (M/s)
1	0.5	0.2	0.25
2	0.6	0.2	0.36

Again, we will apply the rate law to each Experiment:

$$\left.\begin{array}{ll} 0.25 = k0.5^n 0.2^m & \text{(Experiment 1)} \\ 0.36 = k0.6^n 0.2^m & \text{(Experiment 2)} \end{array}\right\} \frac{0.25}{0.36} = \frac{\cancel{k}}{\cancel{k}} \cdot \left(\frac{0.5}{0.6}\right)^n \cdot \left(\frac{\cancel{0.2}}{\cancel{0.2}}\right)^m$$

This approach will cancel k and $[B]$ leaving only n giving

$$0.694 = (0.833)^n$$

In order to solve for n we need to apply natural logarithms (or simply log) to both sides of the equation in order to move the exponent down:

$$Ln(0.694) = n \cdot Ln(0.833) \text{that gives } n = \frac{Ln(0.694)}{Ln(0.833)} = 1.99 = 2$$

Again, remember the orders are normally integer numbers so you can round to the closes integer. In order to calculate the order or B we will select the data with different values of $[B]$ and the same value of $[A]$:

Experiment	$[A]$ (M)	$[B]$ (M)	r (M/s)
1	0.5	0.2	0.25
3	0.5	0.3	0.375

Again, we will apply the rate law to each Experiment:

$$\left.\begin{array}{rl} 0.25 = k0.5^n 0.2^m & \text{(Experiment 1)} \\ 0.375 = k0.5^n 0.3^m & \text{(Experiment 3)} \end{array}\right\} \frac{0.25}{0.375} = \frac{\cancel{k}}{\cancel{k}} \cdot \left(\frac{\cancel{0.5}}{\cancel{0.5}}\right)^n \cdot \left(\frac{0.2}{0.3}\right)^m$$

We have that $0.666 = (0.66)^m$. The order of B is m=1. Now that we have the order, we can use the rate law again to obtain the rate constant using any of the experiments:

$$0.25 = k0.5^2 0.2^1 \quad \text{(Experiment 1)}$$

r=0.5M/s and $[A]$ =0.5M, and given that now we know n=2

$$0.5 = k0.5^2$$

Solving for k we have k=2 1/M^{2}s. We have that the reaction law is: $r = 2[A]^2[B]^1$

Study Check

Use the data below to calculate the rate law of the following reaction: A + B $\longrightarrow$ C

Experiment	$[A]$ (M)	$[B]$ (M)	r (M/s)
1	0.5	0.2	2×10^{-4}
2	0.6	0.2	2×10^{-4}
3	0.5	0.3	4.5×10^{-4}
4	0.6	0.3	4.5×10^{-4}

▶Answer: $r = 5 \times 10^3 [B]^2$

Table 3.2 The integral method

Order	y	x	Slope	Slope Sign	Intersect
0	[A]	t	$-k$	$-$	$[A]_0$
1	$Ln([A])$	t	$-k$	$-$	$Ln([A]_0)$
2	$\frac{1}{[A]}$	t	$+k$	$+$	$\frac{1}{[A]_0}$

`The integral method: an introduction` The integral method is used to calculate the rate law if data reporting concentration and time are given. To understand the use of the integral method, it is convenient to learn more about integral rate laws. Let us analyze the integral rate law for a zeroth-order reaction:

$$[\mathrm{A}] = -k \cdot t + [\mathrm{A}]_0$$

If you look carefully you could notice that this integral rate law is indeed a linear relationship. The term $[\mathrm{A}]_0$ and k are constants. Differently, the terms A] and t are variables. Linear relationships are mathematical relationships with an independent variable represented in the horizontal axes and a dependent variable represented in the vertical axis. In the rate law above the independent variable is t and the dependent variable is $[\mathrm{A}]$ and for every value of t we would have a given value of $[\mathrm{A}]$. Moreover, $-k$ is the slope of the relationship, and $[\mathrm{A}]_0$ is the intersect. Let us analyze another integral rate law, this time for a first-order reaction:

$$\mathrm{Ln}[\mathrm{A}] = -k \cdot t + \mathrm{Ln}[\mathrm{A}]_0$$

we have that this law also represent a linear relationship as well, with t as independent variable (the x) and $Ln([\mathrm{A}])$ as the dependent variable (the y). The slope is $-k$ and the intersect is $Ln([\mathrm{A}]_0)$. Overall, we have that all integral rate laws are linear relationship with t as the x and different properties as the y: $[\mathrm{A}]$ for a zeroth-order, $Ln([\mathrm{A}])$ for a first-order reaction and $\frac{1}{[\mathrm{A}]}$ for a second-order reaction. We also have that the slope of the lines in absolute value always gives k. This information is summarized in Table 3.2.

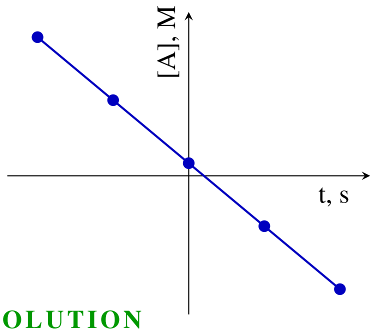
The integral method is based on plotting some kinetic data, for example, concentration vs. time, and fitting the data into a linear relationship. If the quality of the fit is good enough, by reading the slope of the line and the intersect one can compute the rate law that rules the reaction. To associate the experimental data into a line we use linear regression, a statistical tool that gives the equation of the best line that fits the experimental data. At the same time, linear regression gives an estimate of the goodness of the fit through a regression parameter, r or r^2, depending on the statistics software employed. Overall, is good to keep in mind that every plot ($[A]$ vs. t, Ln$[A]$ vs. t or $1/[A]$ vs. t) is associated to a specific order.

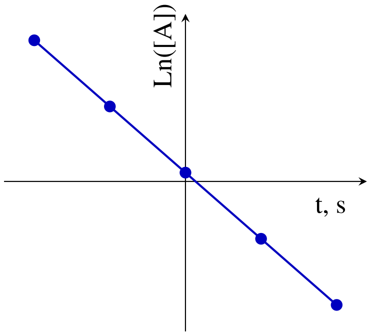
`The integral method: associating a linear plot with an order`

The base of the integral method is to create three different plots: (a) $[\mathrm{A}]$ vs. t (b) Ln$[\mathrm{A}]$ vs. t and (c) $\frac{1}{[\mathrm{A}]}$ vs. t. In all plots, time is always indicated in the horizontal axis. Based on the data, only one of the plots will be the best linear plot with a regression coefficient r^2 closer to one. The perfect linear plot will be associated with an order (here, either 0, 1, or 2). If the plot $[\mathrm{A}]$ vs. t gives a perfect line the reaction will be zeroth-order. If the plot $Ln([\mathrm{A}])$ vs. t gives a perfect line the reaction will be first order. If the plot $\frac{1}{[\mathrm{A}]}$ vs. t gives a perfect line the reaction will be second-order. It is convenient to practice identifying the different plots and associating them with an order. The following example covers just that.

Sample Problem 9

The following plots results from processing kinetic data by means of the integral method. They are all perfect lines with r^2=0.99. Indicate the order of the reaction.



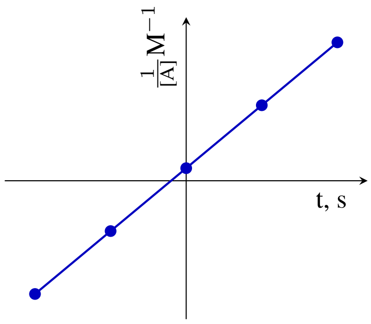


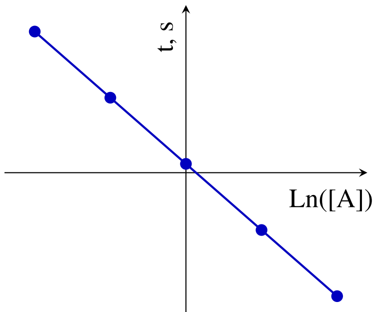
SOLUTION

We will inspect the linear plot and based on what is represented on the vertical and horizontal axis we will conclude the reaction order. The plot on the left represents molarity on the vertical axis and time on the horizontal. This is a zeroth-order reaction. On the other hand, the plot on the right represents the logarithm of molarity on the vertical axis and time on the horizontal axis. This is a first-order reaction.

STUDY CHECK

The following plots results from processing kinetic data by means of the integral method. They are all perfect lines with r^2=0.99. Indicate the order of the reaction.





▶Answer: Left (order 2), right (order 1)

▼Speed is a critical concept in chemistry

▼Explosions are fast reactions

▼Sparks can activate reactions

`How to carry out a linear regression with a calculator` First and foremost, to carry out a linear regression, you need a graphing calculator. In general, Casio calculators are very user-friendly and easy to use. Differently, Texas calculators tend to be hard to employ. Here I will show how to carry out a linear regression displaying the correlation coefficient r^2 parameter on the screen first using a Casio calculator and then using a Texas calculator. You need to follow the following steps to carry out a linear regression in a Casio fx 9750GII:

1. **Step one:** Enter the data: start the calculator and press menu+2 (stat) to access the list menu. You will follow all the steps on this menu. Type the X values in List 1 and Y in List 2.
2. **Step two:** Press Calc (F2) on the bottom menu of the screen, then press Reg (F3) to press X (F1) and then $ax + b$ (F1).
3. **Step three:** At this point, you will have all the results from the regression on the screen.

You need to follow the following steps to carry out a linear regression in a TI 84:

1. **Step one:** Enter the data: press Stat and Enter. Type the X values in List 1 and Y in List 2.
2. **Step two:** Press Second and Zero to access the catalog. Now, go down the menu until you find DiagnosticOn and select it.
3. **Step three:** Now carry out the regression by pressing Stat and Calc. Now go down on the menu until you find $ax + b$ and press it.
4. **Step four:** At this point, you will have all the results from the regression on the screen.

`How to carry out a linear regresion using GraphPad`

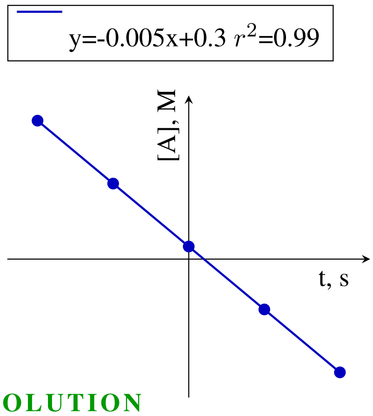
GraphPad.com offers a free online tool to carry out simple linear regressions. Just go to www.graphpad.com/quickcalcs/linear1 and enter the data into the table. Then click on *Calculate Now* to obtain the results. You will find the slope listed and the r^2 listed under the Goodness of fit section. Interestingly, the error in the slope is also listed. Using this simple online tool, you can carry out a linear regression even without a calculator.

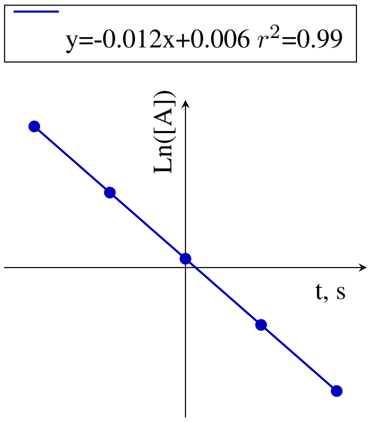
`The integral method: extracting data from linear regressions`

We will practice the integral method by extracting the rate constant and the initial reactant concentration from the linear regression data. Linear regression software will give the equation of a line and the associated r coefficient. We will have to read the slope and the intercept of the equation and infer the rate constant and the initial reactant concentration $[A]_0$. The rate constant will be obtained from the slope of the line, whereas $[A]_0$ is related to the intercept. For all cases, the absolute value of the slope of the line equation will correspond to the rate constant k. The relationship between $[A]_0$ and the intercept differs based on the order. For a zeroth-order plot, the intersect directly corresponds to the initial concentration. Differently, for a first-order reaction, the intercept corresponds to $Ln([\mathrm{A}]_0)$. Therefore, to obtain $[A]_0$ you will have to compute the exponential of the intersect: $[A]_0 = e^{Intersect}$. Finally, for a second-order reaction, the slope will correspond to the inverse of the initial concentration, and hence, to obtain $[A]_0$ we will have to compute the inverse of the intersect: $[A]_0 = \frac{1}{Intersect}$. The following example shows how to obtain the rate constant and $[A]_0$ using regression results.

Sample Problem 10

The following plots results from processing data by means of the integral method. Interpret the linear regressions and indicate the rate law.



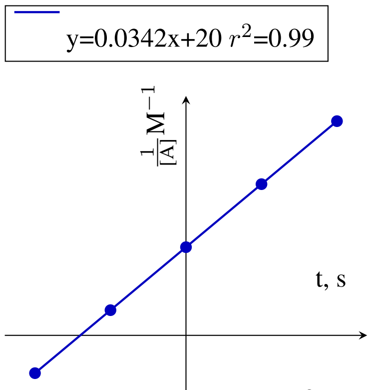


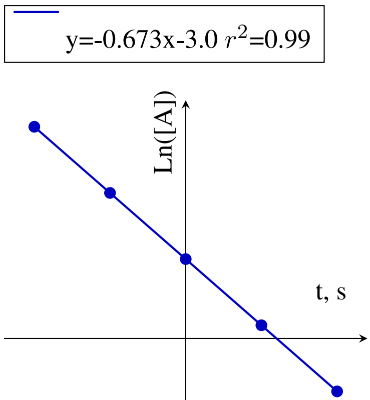
SOLUTION

We will inspect the linear regression to identify the reaction order. From the absolute value of the slope, we will obtain the rate constant. The plot from the left represents molarity and hence this is a zeroth-order reaction with a rate constant k=0.005M/s. The rate law would be: $r = 0.005$. The plot from the right displays the logarithm of the concentration and hence this would be a first-order reaction, with a rate constant k=0.012 1/s. The rate law would be: $r = 0.012$[A]. We can also calculate the initial concentration for each plot. On the left plot, we have that as this is a zeroth-order reaction the intercept directly gives you the initial concentration of reactants: $[A]_0$=0.3M. On the right plot, we have that as this is a first-order reaction the intercept gives you the logarithm of the initial concentration of reactants. We have that $Ln([A]_0)$=0.006 and hence $[A]_0$=$e^{0.006}$=1.006M.

STUDY CHECK

The following plots results from processing data by means of the integral method. Interpret the linear regressions and indicate the rate law and the initial concentration of reactants.





▶Answer: Left ($r = 0.0342[A]^2$, $[A]_0$=0.05M), right ($r = 0.673[A]$, $[A]_0$=0.05M)

Sample Problem 11

Using the following data, apply the integral method to calculate the order and rate constant and write down the rate law.

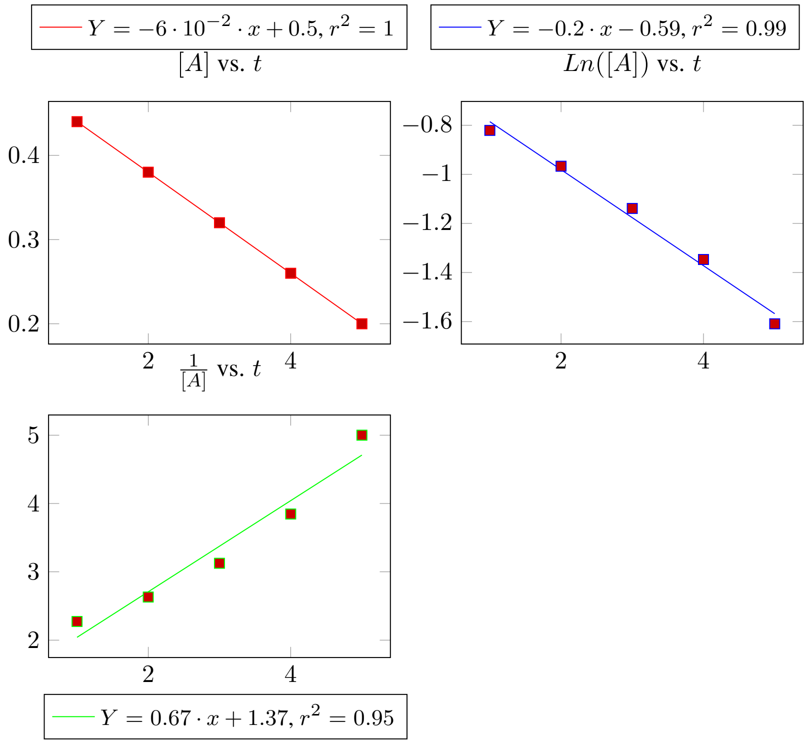
t (s)	1	2	3	4	5
$[A]$, (M)	0.44	0.38	0.32	0.26	0.20

Solution

We will apply the integral method and will process the data calculating the logarithm of the molarity and the inverse of molarity. It is recommended to use three digits in order for the regression to be consistent:

t (s)	$[A]$, (M)	$Ln([A])$	$\frac{1}{[A]}$, (M^{-1})
1	0.440	-0.821	2.273
2	0.380	-0.967	2.631
3	0.320	-1.139	3.125
4	0.260	-1.347	3.846
5	0.200	-1.609	5.000

Now we will create three different plots and carry linear regressions for each:



We have the the reaction can be order zero or order one, as for both cases r^2 is larger or equal to 0.99. However, the fit for zeroth-order is better, so we will choose that possibility. For an oder zero plot, the absolute value of the slope gives the rate constant. The final rate law would be: $r = 6 \times 10^{-2}$.

Study Check

Using the following data, apply the integral method to calculate the order and rate constant and write down the rate law.

t (s)	0	6	12	18	24
$[A]$, (M)	1.0000	0.5000	0.2500	0.1250	0.0625

▶Answer: $r = 0.115[A]$

`The integral method in action` Now we are ready to work from scratch on a set of data from the integral method. Normally, data involving concentration and time will be given. To apply the integral method, we need to process these data into three different columns, reporting the concentration, the logarithm of the concentration, and the inverse of the concentration. Next, we will do three plots and we will obtain the regression line for each plot. We will compare the goodness of the fit by assessing the value of r^2. The closer this value to one the better the fit. This analysis will lead to the reaction order. Finally, after we have selected the plot that corresponds to the best fit, we will analyze the regression line and extract the rate constant and initial molarity of reactants.

`The integral method for more than one reactant` The integral methods can also be applied to obtain rate laws involving more than one reactant (e.g. $r = k[\mathrm{A}][\mathrm{B}]^x$). For this case we will obtain pseudo rate-constants. In general rate constants only depend on the temperature and not on concentration. However, let us assume we want to calculate a rate law that depends on two reactants ([A] and $[\mathrm{B}]^x$), and we start the experiment by using a very large molarity of one of them so that during the reaction this value do not change much. On the other hand, we use a smaller molarity for the other reactant and we want to apply the integral method to his molarity. We will have that [A] is approximately constant (let us call this $[\mathrm{A}]_0$), whereas [B] is not. We can include the constant molarity in the rate law (as it is constant) so that we have

$$r = k[\mathrm{A}]_0[\mathrm{B}]^x \simeq k'[\mathrm{B}]^x$$

And we have that

$$k' = k[\mathrm{A}]_0$$

Applying the integral method we can find out the order of B and the pseudo rate constant. Once we have the pseudo constant, as we control $[\mathrm{A}]_0$ we can always obtain the real rate constant.

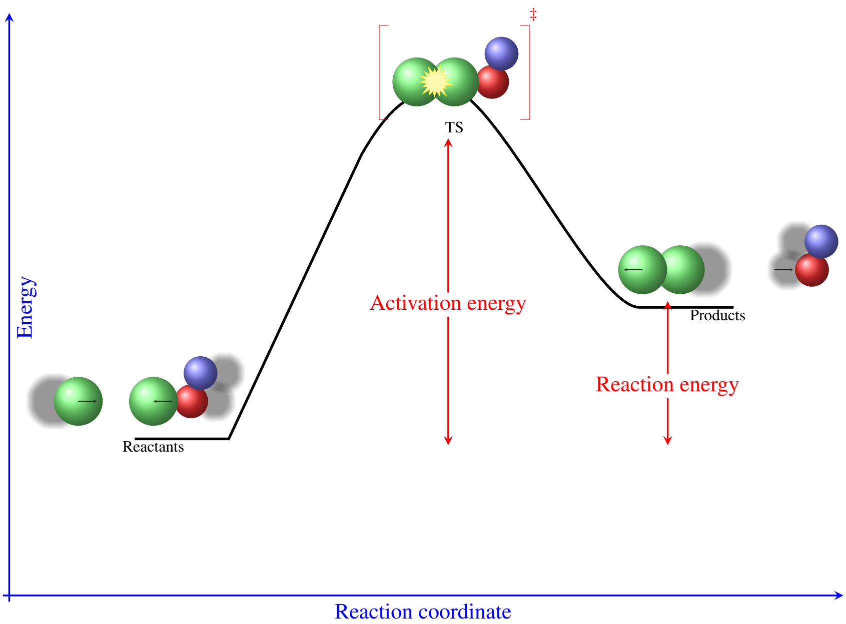


Figure 3.5 Energy diagram showing the conversion of reactants into products

3.5 Collision theory

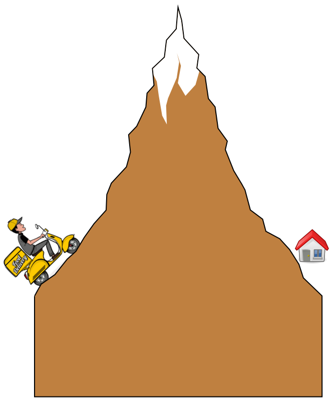
The rate of reaction depends on temperature as reactions proceed at a higher speed at higher temperatures. This is the reason we use refrigerators and freezers to keep food for a longer time. The collision theory provides a microscopic understanding of chemical reactions. A pivotal element of this theory is the transition state also called the activated complex. This is an ephemeral state between reactants and products so that if the reaction proceeds until this point, reactants will certainly evolve into products. The energy difference between the reactants and the transition state of a reaction is called the activation energy. Energy diagrams are useful representations of the energy changes involved in a reaction. By analyzing an energy diagram we can predict whether a reaction is endo of exothermic and estimate the energy involved in the activation of a reaction. This section covers the fundamental principles of collision theory.

`Collision theory` Chemical reactions proceed at a molecular level by means of the collision of reactant molecules. The frequency of collisions is directly related to the reaction rate so that the more collisions per second the faster the reaction would proceed as the higher would be the chance for a reaction to occur. The collision theory of the chemical reactions associates the rate to the frequency of collisions. By increasing the concentration of reactants the number of collisions would increase and that is the reason why there is a concentration dependency on the reaction rate. However, not all collisions are equally effective and only effective collisions will contribute to chemical reactions. Two main factors control the effectiveness of the molecular collisions. First, the orientation of the collisions is critical and only certain orientations will produce successful collisions. The second factor is the energy of the molecules colliding. Only molecules with energy beyond a threshold will be able to form products. In other words: a successful collision should have the right orientation while having enough energy. Let us elaborate on the role of the orientation for the reaction (see Figure 3.6):

$$\mathrm{Cl + NOCl \longrightarrow Cl_2 + NO}$$

We have that a Cl atom reacts with a NOCl molecule to produce Cl_2, as represented in the image below. Only with the Cl atoms hit the NOCl molecule through the Cl atom, the collision will proceed with the right orientation in order to produce the Cl_2 molecule. At the same time, if the orientation is correct and the molecules have enough energy a transition state will form. We will discuss more this state during the next sections.

`Transition state and activation energy` We have discussed that the energy of reactants is a key parameter during a reaction. The minimum energy needed to activate the reactants so that they can produce products is called activation energy. The higher the activation energy the more energy is needed for a reaction to happen and hence the lower the reaction rate will be. When reactants get together they increase their energy from reactants until a state of maximum energy before producing the products. This state of maximum energy is called an activated complex or transition state. Transition states are very ephemeral states that exist for a very short amount of time in comparison to reactants and products. An example of activation happens during the ignition of a gas burner. Cooking gas and oxygen can react together to produce water and carbon dioxide only with the help of a spark. A spark is needed to activate the reactants so that their molecules can reach the transition state and hence produce products. Transition states are normally indicated with a ‡ sign to indicate they have a very different nature than normal molecules. The figure below represents two reaction pathways with different activation energy.



`Effect of temperature` As temperature increases the kinetic energy of the molecules reacting increases as well and with this the energy involved in the collisions also increases. Molecules move faster at higher temperature and hence the frequency of collisions between reactants and molecules increases with temperature. Overall, the reaction rate increases with temperature. In other words, we can speed up chemical reactions by increasing temperature.

`Concentration of reactants` Another way to favor the collision between the molecules of reactants is to increase its concentration. The larger the concentration of reactants the higher the probability of their molecules to collide with each other potentially producing products.

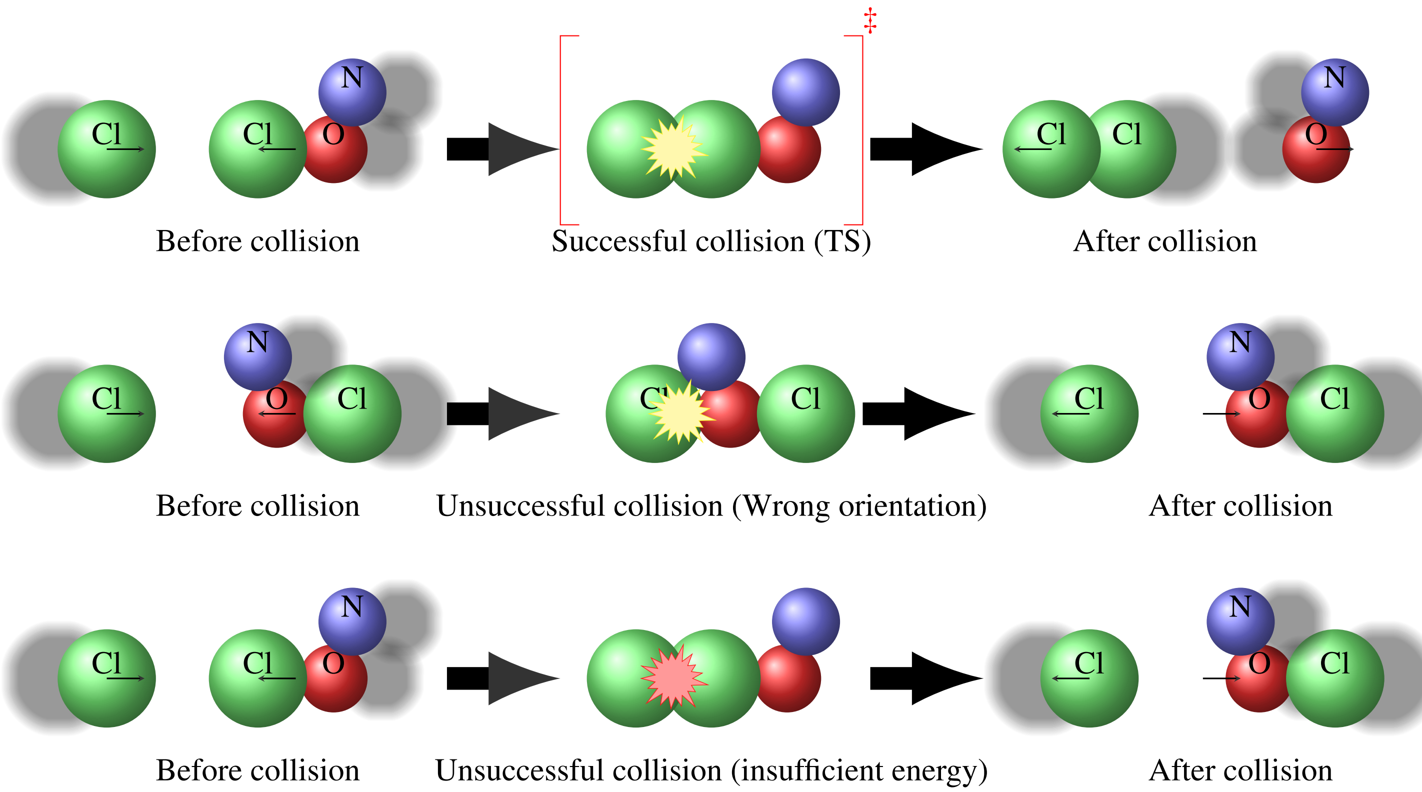


Figure 3.6 Example of three collisions, a successful collision that leads to a transition state and the products and two unsuccessful collisions due to an ineffective molecular arrangement and insufficient energy.

Sample Problem 12

Methane (CH_4) reacts with oxygen (O_2) to produce carbon dioxide (CO_2) and water (H_2O) according to the following reaction

$$CH_4(g) + O_2(g) \longrightarrow CO_2(g) + H_2O(g)$$

(a) What happens to the number of collision between CH_4 and O_2 when you add extra oxygen? (b) How would increasing temperature impacts the rate of the reaction?

Solution

(a) The more reactants the mode collisions between their molecules. (b) Increasing temperature would increase the rate of the combustion reaction, as increasing temperature increases collisions.

Study Check

How would the following changes affect the rate of this reaction:

$$2\,H_2(g) + O_2(g) \longrightarrow 2\,H_2O(g)$$

(a) Removing oxygen; (b) Decreasing temperature

Answer: (a) slow down; (b) slow down.

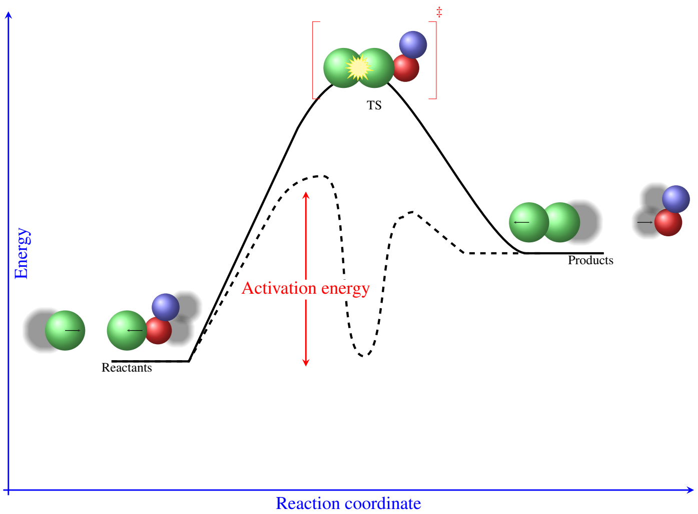


Figure 3.7 Energy diagram showing the catalyzed and non-catalyzed pathways

`Catalysts` Catalysts are chemicals that are not directly involved in a reaction–they are neither reactants nor products and as such, they are not consumed during a reaction. Still, they have the ability to increase the rate of reactions. Each catalyst is specific to a given reaction. As such, a catalyst can speed up a reaction without affecting other reactions. Enzymes are biological catalysts that support chemical reactions happening in the body. Most of the biological reactions involved in the chemistry of life are very slow. It is only with the help of biological catalysts that these reactions can indeed happen at reasonable rates. Every single metabolic reaction has associated an enzyme responsible for speeding up the process. Similarly, catalysts are well-known compounds in the chemical industry. The production of chemicals in the industry is also supported by the help of catalysts which make the chemical processes more economically beneficial. For example, catalysts are used in the manufacturing of margarine from vegetable oils.

In the same line, the synthesis of ammonia is catalyzed by iron, the manufacturing of sulfuric acid uses catalysts based on nitrogen(II) oxide and platinum, and the oxidation of hydrocarbons in automobile exhausts are catalyzed by platinum. Enzymes are also added to detergents in order to speed up the breaking of protein molecules in stains. How do catalysts work? Catalyst function by providing alternative reaction pathways with more favorable–normally lower–activation energy. This way, in the presence of a catalyst, reactions proceed by means of two different reaction pathways, the catalyzed and non-catalyzed pathways. Still, as the catalyzed pathway is more favorable the overall speed of the reaction is higher. Catalysts can be classified as homogeneous, heterogeneous, and acid catalysts.

`Heterogeneous catalysis` A heterogeneous catalyst is a material with catalytic properties which has a different state of matter than the main reaction. For example, the synthesis of ammonia is carried in the chemical industry by means of reacting has hydrogen and nitrogen. The catalyst for this process is iron, which is a solid. The reaction proceeds in a gas phase, whereas the catalyst employed is solid. This is an example of heterogeneous catalysis. Another example of the hydrogenation of alkenes by means of molecular hydrogen. This process is used for example in the production of saturated fats or shortenings such as Crisco. As hydrogen hardly dissociates in gas, this reaction proceeds at a very low rate without the help of a catalyst. Differently, in the presence of platinum, the dissociation energy of hydrogen is lower what makes the process function at a faster rate. The necessary steps for a reaction to happen on a heterogeneous catalyst are: (a) the adsorption of the reactants on the surface (the words adsorption refers to the attachment of a gas onto a solid) (b) the activation of the reactants in which its molecular geometry changes due to the adsorption (c) the migration of the adsorbed reactants (d) the surface reaction (e) the desorption of the products

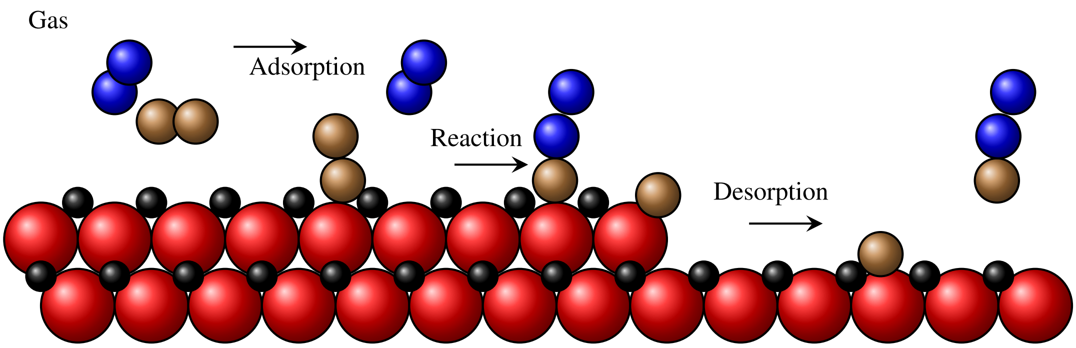
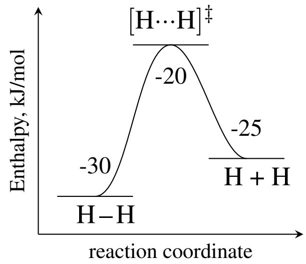


Figure 3.8 Steps in a heterogeneously catalyzed reaction

3.6 Use of energy diagrams

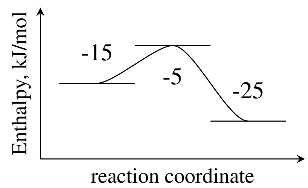
`Energy diagrams` Energy profiles, or energy diagrams, are a visual representation of the advancing of a reaction (see Figure 3.5). Reactions advance through changes in the geometry of the reaction and during a reaction, for example, some bonds are being formed while others are being broken. The horizontal axis of the diagram represents the reaction coordinate, representing how the reaction proceeds, from reactants on the left to products on the right. The vertical axis represents the energy, oftentimes the enthalpy of the different steps of the reaction. Transitions states are maxima in the energy diagrams, that is, points between two minima connecting reactants and products.



We can extract two types pf properties from energy diagrams. First, the reaction energy ΔH_R is the difference in energy between the products and reactants. If this value is negative, the reaction will be exothermic, releasing heat. The opposite is true for an endothermic reaction that consumes heat. Second, we can obtain the activation energy as the energy of the transition state with respect to the energy of the reactants. The activation energy will always be a positive number which value is related with the reaction rate: in general small activation energies lead to fast reactions and large activation energies lead to slow reactions. Below are two examples showing an exothermic and endothermic reactions, as well as a diagram showing the activation energy.

Sample Problem 13

For the energy profile below



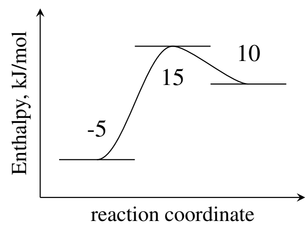
Calculate (a) The energy of the reactants (b) The energy of the products (c) The energy of the transition state (d) The activation energy (e) The reaction energy (f) Indicate whether the reaction is endothermic or exothermic

SOLUTION

We have that the energy of the reactants is -15kJ/mol, the energy of the products is -25kJ/mol and the energy of the transition state (TS) is -5kJ/mol. The reaction energy represents the energy of the products concerning the reactants: $-25-(-15) = -10$kJ/mol. This is an exothermic reaction. The activation energy represents the energy of the TS concerning the reactants: $-5-(-15) = 10$kJ/mol.

STUDY CHECK

For the energy profile below



Calculate (a) The energy of the reactants (b) The energy of the products (c) The energy of the transition state (d) The activation energy (e) The reaction energy (f) Indicate whether the reaction is endothermic or exothermic

▶Answer: (a) -5kJ/mol (b) 10kJ/mol (c) 15kJ/mol (d) 20kJ/mol (e) 15kJ/mol (f) Endothermic

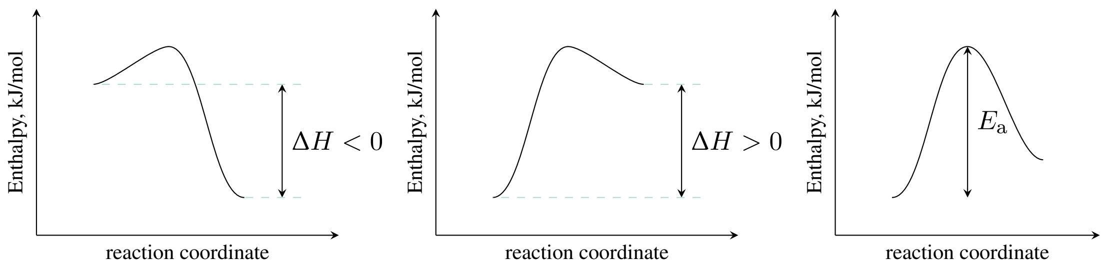


Figure 3.9 A set of reaction energy profiles displaying the energy of reaction and the activation energy: exothermic (left), endothermic (center), and endothermic (right)

3.7 The Arrhenius equation

The Arrhenius equation on its exponential form The Arrhenius equation gives the dependency of the rate constant with temperature. We will introduce this equation in two forms: an exponential and a linear form. However, in this section we will focus on the exponential form of this equation:

$$\boxed{k = Ae^{\left(-\frac{E_a}{RT}\right)}} \text{ and } \boxed{Ln(k) = Ln(A) - \frac{E_a}{R} \cdot \left(\frac{1}{T}\right)} \tag{3.4}$$

where:

r is the overall rate of reaction

A is the frequency (or pre-exponential) factor

E_a is the activation energy in kJ/mol

R is the constant of the gases in energy units (8.314J/molK)

T is the temperature in Kelvins

Based on Arrhenius equation, the larger the activation energy the smaller the rate constant. Similarly, as the temperature increases the rate constant increases, and therefore the reaction rate increases too. The following example shows how to use this equation in its exponential form. The linear form of the equation results simply from applying logarithms to both sides of the regular Arrhenius equation and is useful to obtain data graphically as it represents a linear relationship with $\frac{1}{T}$ as x and $Ln(k)$ as y. The slope of the resulting line is $-\frac{E_a}{R}$ (mind the negative sign) and the intercept is $Ln(A)$. In the Arrhenius equation, the pre-exponential factor represents the frequency of collision between molecules, in s^{-1}.

Sample Problem 14

The rate constant at 325K for the decomposition reaction $C_4H_8 \longrightarrow 2\,C_2H_4$? is $6.1 \times 10^{-8}\ s^{-1}$, and the frequency factor is $3.9 \times 10^{-5}\ s^{-1}$. Calculate: (a) the activation energy (b) the rate constant at 200K

SOLUTION

We will use the Arrhenius equation on its exponential form:

$$k = Ae^{\left(-\frac{E_a}{RT}\right)}$$

As we know the frequency factor ($A = 3.9 \times 10^{-5}$ s^{-1}) and the rate constant at a given temperature ($k = 6.1 \times 10^{-8}$ s^{-1} at 325K), we can solve for the activation energy:

$$6.1 \times 10^{-8} = 3.9 \times 10^{-5} e^{\left(-\frac{E_a}{8.314 \cdot 325}\right)}$$

We have that

$$1.56 \times 10^{-3} = e^{\left(-\frac{E_a}{8.314 \cdot 325}\right)}$$

Using logarithms to solve for E_a:

$$Ln(1.56 \times 10^{-3}) = -\frac{E_a}{8.314 \cdot 325}$$

And solving for E_a:

$$Ln(1.56 \times 10^{-3}) = -\frac{E_a}{8.314 \cdot 325}$$

we have that E_a=17463.5J/mol in another words17kJ/mol.

STUDY CHECK

The Arrhenius parameters for the gas-phase reaction below are 6×10^{-12} s^{-1} and 665J/mol.

$$H_2S + OH \longrightarrow HS + H_2O$$

Calculate: (a) the rate constant at 300K. (b) the temperature in Kelvins at which the rate constant is 5×10^{-12} s^{-1}

Answer: (a) 4.6×10^{-12} s^{-1} (b) 439K

The following example shows how to use rate constant and temperature data in order to graphically compute the Arrhenius parameters.

Sample Problem 15

Using the following data, calculate the Arrhenius parameters (the activation energy and the frequency factor) for the following gas-phase reaction using the graphic method:

$$HS + O_3 \longrightarrow HSO + O_2$$

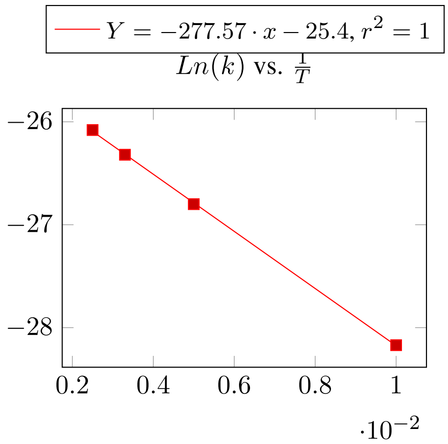
T (K)	k, (s^{-1})
100	5.8×10^{-13}
200	2.3×10^{-12}
300	3.7×10^{-12}
400	4.7×10^{-12}

Solution

As they provide with rate constant and temperature data, we will use the linearized form of Arrhenius equation in order to calculate the activation energy and the frequency factor. We will first process the data and compute the natural logarithm of the rate constant and the inverse of temperature.

T (K)	$\frac{1}{T}$ (K^{-1})	k, (s^{-1})	$Ln(k)$
100	1.0×10^{-2}	5.8×10^{-13}	-28.17
200	5.0×10^{-3}	2.3×10^{-12}	-26.80
300	3.3×10^{-3}	3.7×10^{-12}	-26.32
400	2.5×10^{-3}	4.7×10^{-12}	-26.08

Now we are ready to graph the Arrhenius plot with $\frac{1}{T}$ in the horizontal (x) axis and $Ln(k)$ in the vertical (y) axis.



From the slope of the graph we will calculate $-\frac{E_a}{R}$ and from the intersect we will calculate $Ln(A)$. We have that $\frac{E_a}{R} = 227.57$ and $Ln(A) = -25.4$. We have that the Arrhenius parameters are: $E_a = 1892 J/mol$ and $A = 9.3 \times 10^{-12}$ s^{-1}.

Study Check

Using the following data, calculate the Arrhenius parameters (the activation energy and the frequency factor) for the following gas-phase reaction using the graphic method:

$$HS + Cl_2 \longrightarrow ClSH + Cl$$

T (K)	k, (s^{-1})
100	1.1×10^{-15}
200	1.4×10^{-13}
300	6.9×10^{-13}
400	1.5×10^{-12}

▶Answer: $E_a = 7990 J/mol$, $A = 1.7 \times 10^{-11}$ s^{-1}.

3.8 Reaction mechanisms

Most chemical reactions do not occur in a single step but in a series of steps that connect reactants with products in the form of a reaction mechanism. Some of the steps of the mechanism are slow–heavily impacting the experimentally measured rate law–others proceed quickly or are equilibrated, hence going forward and backward at the same rate. A proper reaction mechanism will justify the experimental rate law while matching the overall reaction stoichiometry. At the same time, experimentally, one observes apparent reaction rate and activation energy, resulting from a combination of rates and activation energies, respectively, for the mechanism. In other words, the reaction rate measured is a combination of the reaction rates of the mechanism behind, and the measured activation energy is a combination of the activation energies of the mechanism behind, as well. This section will provide insight into the different nuances of reaction mechanisms, beginning with an example of a reaction mechanism, followed by a description of two of the most important steps in mechanisms: the rate-limiting step and the equilibrated steps. The first step provides the experimental rate law, whereas the second is useful for calculating the concentration of reaction intermediates.

`An example of a reaction mechanism` Let us analyze the reaction between nitrogen dioxide and carbon monoxide, with the experimental rate law indicate below:

$$NO_{2(g)} + CO_{(g)} \longrightarrow NO_{(g)} + CO_{2(g)} \qquad r^{Exp} = k^{Exp}[NO_2]^2$$

The experimental rate law demonstrates that this is a second order reaction. The reaction mechanism associated with the overall reaction is shown below.

$$(1) \qquad NO_{2(g)} + NO_{2(g)} \xrightarrow{k_1} NO_{3(g)} + NO_{(g)} \qquad \text{(slow)} \qquad r_1 = k_1[NO_2]^2$$
$$(2) \qquad NO_{3(g)} + CO_{(g)} \xrightarrow{k_2} NO_{2(g)} + CO_{2(g)} \qquad \text{(fast)} \qquad r_2 = k_2[NO_3][CO]$$

It consists of two steps. The first step involve the reaction of two nitrogen dioxide molecules (NO_2)–one of the reactants–to produce a molecule of nitrogen trioxide and a nitrogen monoxide molecule (NO_3)–one of the products. The second step involves the reaction of nitrogen trioxide (NO_3) and carbon monoxide (CO)–a second reactant–to produce nitrogen dioxide and carbon dioxide–the second product. All reactions in a mechanism are called *elemental steps* and hence they proceed in a single step. At the same time, for each step, the number of molecules involved in the reactants is called *molecularity* and directly corresponds to the order or the rate law for a given step. Reactions involving a single molecule are called unimolecular, whereas reactions involving two molecules are called bimolecular. Reactions involving three molecules are called termolecular and are very rare. We have that every single step of the mechanism has a reaction constant (k_1 and k_2) and a rate law (r_1 and r_2). The rate law of each step of the mechanism simply results from multiplying the rate constant by the molarity of all reactants, using the stoichiometric coefficients are powers. For example the first step involves two NO_2 molecules and hence the rate law involve $[NO_2]^2$. At the same time, every step is characterized by a certain speed. Some steps are slower (step 1) than other (step 2). By combining the different steps and eliminating the repeated species we can obtain the overall reaction:

$$(1) \qquad NO_{2(g)} + NO_{2(g)} \xrightarrow{k_1} NO_{3(g)} + NO_{(g)} \qquad \text{(slow)} \qquad r_1 = k_1[NO_2]^2$$
$$(2) \qquad NO_{3(g)} + CO_{(g)} \xrightarrow{k_2} NO_{2(g)} + CO_{2(g)} \qquad \text{(fast)} \qquad r_2 = k_2[NO_3][CO]$$

$$(1)+(2) \quad NO_{2(g)} + \cancel{NO_{2(g)}} + \cancel{NO_{3(g)}} + CO_{(g)} \longrightarrow \cancel{NO_{3(g)}} + NO_{(g)} + \cancel{NO_{2(g)}} + CO_{2(g)}$$
$$r^{Exp} = k^{Exp}[NO_2]^2$$

Finally, there are four different types of molecules in a mechanism: overall reactants, overall products, intermediates, and catalysts. Overall, reactants are the reactants of the overall reaction (indicated in green). Overall products are the products of the overall reaction (indicated in orange). Intermediates (indicated in red) are generated only inside the mechanism and are not involved in the overall reaction. They are very ephemeral states that exist only temporally, and they are never part of any experimental rate law. More importantly, their concentration does not increase or decrease progressively with time and can often increase to then decrease. Finally, catalysts are species involved in the reaction that remain unaltered (they appear as products and as reactants).

`Rate limiting step` The rate limiting step of a mechanism approximately corresponds to the slowest step as this step will determine the experimental rate law:

$$\boxed{r^{Exp} \sim r^{slow}} \tag{3.5}$$

The sign $\sim$ indicated that this is an approximation. Rate limiting steps represent bottlenecks of chemical mechanisms limiting the overall advance of the reaction. Hence the rate liming step is crucial to obtain the overall experimental rate law.

`Equilibrated steps` Often times we encounter equilibrated steps in mechanisms. Let us analyze the decomposition of ozone to produce molecular oxygen:

$$2\,\mathrm{O}_{3(g)} \longrightarrow 3\,\mathrm{O}_{2(g)} \qquad r^{Exp} = k^{Exp}\frac{[\mathrm{O}_3]^2}{[\mathrm{O}_2]}$$

This reaction proceed by means of the following mechanism the steps of which combined gives the overall reaction:

(1) $$\mathrm{O}_{3(g)} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{O}_{2(g)} + \mathrm{O}_{(g)} \qquad 0 = k_1[\mathrm{O}_3] - k_{-1}[\mathrm{O}_2][\mathrm{O}]$$

(2) $$\mathrm{O}_{(g)} + \mathrm{O}_{3(g)} \xrightarrow{k_2} 2\,\mathrm{O}_{2(g)} \qquad \text{(slow)} \qquad r_2 = k_2[\mathrm{O}][\mathrm{O}_3]$$

(1)+(2) $$\mathrm{O}_{3(g)} + \cancel{\mathrm{O}_{(g)}} + \mathrm{O}_{3(g)} \longrightarrow \mathrm{O}_{2(g)} + \cancel{\mathrm{O}_{(g)}} + 2\,\mathrm{O}_{2(g)}$$
$$r^{Exp} = k^{Exp}\frac{[\mathrm{O}_3]^2}{[\mathrm{O}_2]}$$

The reactant is O_3, whereas O_2 is the product and O is an intermediate. The first step, the step indicated with a double arrow, is equilibrated, that means that the forward reaction ($\longrightarrow$) proceeds fast enough in order to be able to proceed backwards ($\longleftarrow$). At this point the forwards reaction is characterize by a rate constant k_1 and the backwards by another rate constant k_{-1}. Equilibrated steps are employed to compute the concentration of the reaction intermediates. As both reactions are equilibrated we have that the forwards and backwards reaction proceed at the same speed giving: $0 = k_1[\mathrm{O}_3] - k_{-1}[\mathrm{O}_2][\mathrm{O}]$. We can compute the concentration of the intermediate by means of the rate law of the equilibrated step:

$$0 = k_1[\mathrm{O}_3] - k_{-1}[\mathrm{O}_2][\mathrm{O}] \Longrightarrow [\mathrm{O}] = \frac{k_1[\mathrm{O}_3]}{k_{-1}[\mathrm{O}_2]}$$

Now let us calculate the expression of the experimental rate law in terms of the different mechanistic steps. The second step is the rate limiting reaction so we have that:

$$r^{Exp} \sim r^{slow} = k_2[\mathrm{O}][\mathrm{O}_3]$$

Using the concentration of the intermediate recently calculates we have:

$$r^{Exp} = \frac{k_2 \cdot k_1}{k_{-1}}\frac{[\mathrm{O}_3]^2}{[\mathrm{O}_2]}$$

We have that the mechanism justifies the experimental rate law with

$$k^{Exp} = \frac{k_2 \cdot k_1}{k_{-1}}$$

The measured reaction constant does not correspond to the constant of any of the steps involved in the mechanism, but a combination of these. For this reason, we call this reaction constant the *apparent* reaction constant.

`Apparent activation energy` We have said that often the measured reaction constant corresponds to a combination of the reaction constants of the steps in the mechanism. For this reason, this constant is referred to as apparent. In a similar way, the measured activation energy often does not correspond to the activation energy of any of the steps, but a combination of these. This activation energy is called *apparent activation energy*. We can compute the apparent activation energy by means of some algebra once we know the expression for the apparent reaction constant. For example, for the case we are studying:

$$k^{Exp} = \frac{k_2 \cdot k_1}{k_{-1}}$$

we need the corresponding activation energies for the steps E_1, E_{-1} and E_2. The rule is to multiply each activation energy by the value of its power in the k^{Exp} expression and add all values. For the example above the apparent activation energy would be?

$$E^{Exp} = E_2 + E_1 - E_{-1}$$

Another example, for the apparent constant:

$$k^{Exp} = \frac{k_2 \cdot k_1}{k_{-1}^2}$$

the apparent activation energy would be

$$E^{Exp} = E_2 + E_1 - 2 \cdot E_{-1}$$

This procedure assumes that the preexponential constant is the same for each step, which comes from the fact that the powers of exponential functions–remember each reaction constant is given by the exponential Arrhenius's la–combine with each other, and for example

$$k^{Exp} = \frac{k_2 \cdot k_1}{k_{-1}} = \frac{A e^{\left(-\frac{E_{a_2}}{RT}\right)} \cdot A e^{\left(-\frac{E_{a_1}}{RT}\right)}}{A e^{\left(-\frac{E_{a_{-1}}}{RT}\right)}} = e^{\left(-\frac{E_{a_2}+E_{a_1}-E_{a_{-1}}}{RT}\right)}$$

`How to verify a possible mechanism` Imagine we suggest a possible mechanism for a reaction with an experimental rate law and we want to make sure this is an appropriate mechanism. An appropriate chemical mechanism follows two rules: (a) the combination of all steps gives the overall reaction and (b) the experimental rate law can be justified by means of the mechanistic steps.

Sample Problem 16

Assess whether the reaction below:

$$2\,NO_{(g)} + Cl_{2(g)} \longrightarrow 3\,NOCl_{(g)} \qquad r^{Exp} = k^{Exp}[NO]^2[Cl_2]$$

is correctly represented by the following mechanism:

$$(1) \qquad NO_{(g)} + Cl_{2(g)} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} NOCl_{2(g)} \qquad \text{(equil.)}$$

$$(2) \qquad NOCl_{2(g)} + NO_{(g)} \xrightarrow{k_2} 2\,NOCl_{(g)} \qquad \text{(slow)}$$

Solution

We will verify two criteria: that the combination of the mechanistic steps gives the overall reaction and that the rate law is correctly described by the mechanism, using the rate determining step approximation–this means that the slow step determines the experimental rate law. We will first write down the rate laws for each individual step and we will combine both steps:

$$(1) \qquad NO_{(g)} + Cl_{2(g)} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} NOCl_{2(g)} \qquad \text{(equil.)}$$

$$0 = k_1[NO][Cl_2] - k_{-1}[NOCl_2][Cl_2]$$

$$(2) \qquad NOCl_{2(g)} + NO_{(g)} \xrightarrow{k_2} 2\,NOCl_{(g)} \qquad \text{(slow)}$$

$$r_2 = k_2[NOCl_2][NO]$$

$$(1)+(2) \qquad NO_{(g)} + Cl_{2(g)} + \cancel{NOCl_{2(g)}} + NO_{(g)} \longrightarrow \cancel{NOCl_{2(g)}} + 2\,NOCl_{(g)}$$

$$r^{Exp} = k^{Exp}[NO]^2[Cl_2]$$

We have that the first criteria is met as the overall reaction is obtained by combining both steps. Now we will use the rate liming approach in order to reproduce the experimental rate law:

$$r^{Exp} \sim r^{slow} = k_2[NOCl_2][NO]$$

and we will use the equilibrated step in order to calculate $[NOCl_2]$:

$$0 = k_1[NO][Cl_2] - k_{-1}[NOCl_2][Cl_2] \Longrightarrow [NOCl_2] = \frac{k_1}{k_{-1}}[NO][Cl_2]$$

Now we are able to compute r^{slow}:

$$r^{slow} = \frac{k_2 \cdot k_1}{k_{-1}}[NO]^2[Cl_2]$$

We have that the rate limiting step gives the experimental rate law. Hence this is an appropriated mechanism with $k^{Exp} = \frac{k_2 \cdot k_1}{k_{-1}}$.

Study Check

Assess whether the reaction below:

$$2\,NO_{(g)} + O_{2(g)} \longrightarrow 2\,NO_{2(g)} \qquad r^{Exp} = k^{Exp}[NO]^2[O_2]$$

is correctly represented by the following mechanism:

$$(1) \qquad NO_{(g)} + NO_{(g)} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} N_2O_{2(g)} \qquad \text{(equil.)}$$

$$(2) \qquad N_2O_{2(g)} + O_{2(g)} \xrightarrow{k_2} 2\,NO_{2(g)} \qquad \text{(slow)}$$

▶Answer: yes with $k^{Exp} = \frac{k_2 \cdot k_1}{k_{-1}}$

3.9 Catalysis by enzymes

Enzymes are biological catalysts responsible for speeding up the rate of biochemical reactions. Two examples of enzymes are *catalase* or *amylase*; the former is responsible for breaking down hydrogen peroxide whereas the latter catalyzes the hydrolysis of starch. Enzymes are particularly relevant in biochemistry accelerating chemical reactions at rates necessary for life, as most biochemical reactions require an enzyme. Here we cover the chemical principles behind biological catalysis, the rate laws that define catalyzed rates, and the factors that impact enzyme activity.

`What are enzymes` Enzymes are large biological molecules, in general proteins, able to speed up chemical reaction. Enzymes are more efficient than regular catalysts. The activation energy of enzyme-catalyzed chemical reactions tend to be smaller than that of regularly-catalyzed reactions. For example, the activation energy of the Fe^{2+}-catalyzed decomposition of hydrogen peroxyde is 42 KJ/mol, whereas that for the catalase-catalyzed decomposition is only 7 KJ/mol.

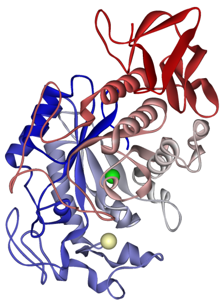
$$H_2O_2 \xrightarrow{\text{catalase}} 2\,H_2O + O_2$$

Enzymes are large molecules, and only a small part of the molecyle called the *active site* is involved in the catalytic process. Enzymes are associated with coenzymes which are inorganic compounds essential for catalyst action. There ara specific and nonspecific enzymes. On one hand, we can find specific catalysts. An example would be urease which only catalyzes the hydrolysis of urea ($CO(NH_2)_2$), the reaction between urea and water:

$$CO(NH_2)_2 + H_2O \xrightarrow{\text{urease}} CO_2 + 2\,NH_3$$

On the other hand, we can also find less specific enzymes such as proteolytic enzymes which catalyze the hydrolysis of the peptide linkage that connects aminoacids in proteins. In fact, these types of enzymes are group-specific and will catalyze the hydrolysis of the peptide linkage in any molecule with such a chemical bond.

▼Amylase



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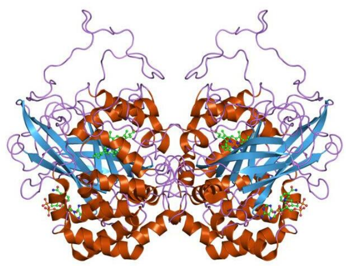
▼Catalase



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`Amylase` Amylase is a digestive enzyme involved in the hydrolysis of complex carbohydrates (like starch) into simpler sugars.

$$\text{starch} \xrightarrow{\text{amylase}} \text{glucose+maltose}$$

It is primarily produced by the pancreas and salivary glands aiding digestion. It is present in human saliva, and it was named after the Ancient Greek name for saliva. It exists in three different forms: $\alpha-$amylase, $\beta-$amylase, and $\gamma-$amylase. $\alpha-$amylase is the major form of amylase found in humans and other mammals, also being the fastest. $\beta-$amylase is present in plants and microbes, being responsible for the ripening of fruit as it breaks starch into maltose, resulting in the sweet flavor of ripe fruit. $\gamma-$amylase, present in animals and microbes, is most active around pH 3, being the most acidic optimum pH of all amylases.

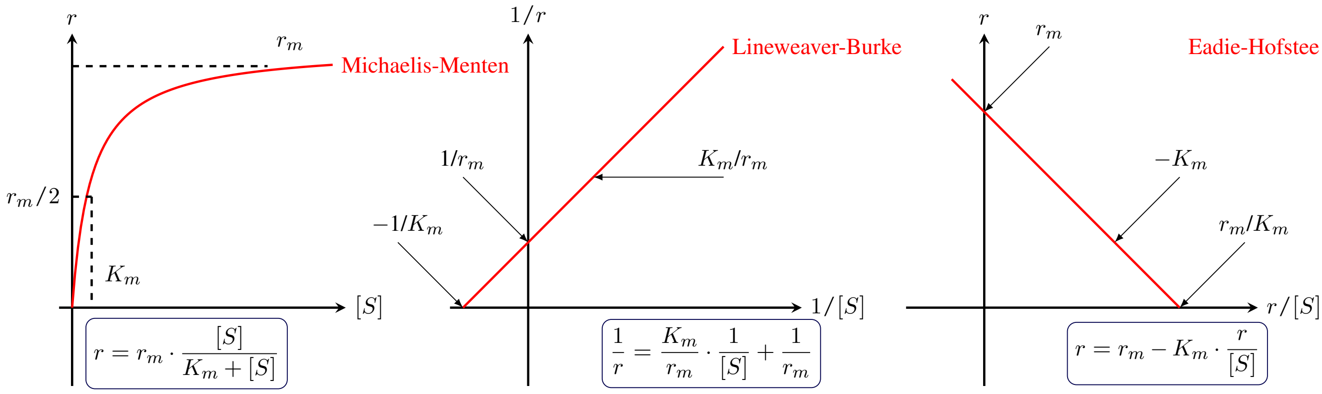


Figure 3.12 Schematic representation of three enzyme kinetic plots. (left) The Michaelis-Menten, hyperbolic plot where rate r is plotted against substrate concentration $[S]$; (center) The Lineweaver-Burk linear plot where the inverse of rate is plotted against inverse substrate concentration giving a positive slope; (right) Eadie-Hofstee plot where the of rate is plotted against the rate between rate and substrate concentration giving a negative slope.

`Michaelis-Menten equation` The Michaelis-Menten equation gives the rate of an enzyme-catalyzed reaction in terms of the concentration of the enzyme.

$$\boxed{r = r_m \cdot \frac{[S]}{K_m + [S]}} \tag{3.6}$$

where:

r is the enzyme-catalyzed reaction rate

$[S]$ is the enzyme concentration

r_m is the maximal velocity

K_m Michaelis constant

It involves two different constants r_m which is the maximal rate and K_m, the Michaelis constant. Enzymes speed up the chemical rate. K_m is a constant that only involves the reaction rate constant of the enzyme-catalyzed reaction mechanism. r_m also depends on the reaction rate constant of the enzyme-catalyzed reaction mechanism, but it also depends on the enzyme concentration. However, when the enzyme concentration is too large, the enzyme-catalyzed rate reaches a plateau with a maximal rate. At this point increasing the enzyme concentration does not increase the reaction rate (see Figure 3.12 (left)). This is because all active sites in the catalyst are occupied by substrate molecules and we say the enzyme reaches *saturaction*. An interesting point is that the Michaelis-Menten rate law for an enzyme-catalyzed reaction does not have a fixed reaction order. In fact, the order changes with substrate concentration. At low concentrations, the reaction is order one, whereas at large concentrations the reaction is order zero.

`Lineweaver-Burk linear equation` Linear relations are suitable for obtaining kinetic data because they are simple to analyze when dealing with large datasets.

Perhaps more importantly, errors are well-understood in linear expressions. The quality of the linear fit simply relates to the correlation coefficient r^2. Unfortunately, the Michaelis-Menten equation is not a linear relationship and hence it is not suitable to fit experimental data. Fortunately, the Michaelis-Menten equation can be recast into a first linear expression by simply using reciprocals. The resulting equation in which the inverse of the rate $1/r$ (vertical axis) is plotted against the inverse of $1/[S]$ (horizontal axis) is called the Lineweaver-Burk equation (see Figure 3.12 (center)).

$$\boxed{\frac{1}{r} = \frac{K_m}{r_m} \cdot \frac{1}{[S]} + \frac{1}{r_m}} \tag{3.7}$$

where:

$1/r$ is the inverse of the enzyme-catalyzed reaction rate

$1/[S]$ is the inverse of the enzyme concentration

r_m is the maximal velocity

K_m Michaelis constant

From the intersept, we can obtain V_m and from the slope, which is positive, we can calculate K_m.

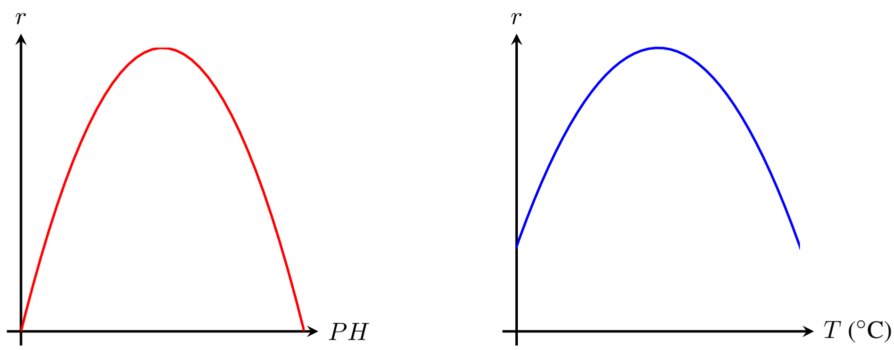


Figure ?? Schematic representation of PH (left) and temperature (right) volcano plots for the activity of amylase. Amylase is the digestive enzyme that breaks down complex carbohydrates (like starch) into simpler sugars, aiding digestion. The optimum temperature is 37°C, and the optimum PH range is 6.7-7.0.

`Eadie-Hofstee linear equation` The Michaelis-Menten equation can be recast into a second linear expression as show below:

$$\boxed{r = r_m - K_m \cdot \frac{r}{[S]}} \tag{3.8}$$

is called the Eadie-Hofstee equation (see Figure 3.12 (right)). In the Eadie-Hofstee equation, the rate $1/r$ (vertical axis) is plot againts $r/[S]$ (horizontal axis). The slope of this plot is negative. The advantage of this equation is that is spreads more the data than the Lineweaver-Burk equation.

Sample Problem 17

ONPG is a synthetic lactose analog used to detect and quantify the activity of the enzyme β-galactosidase. The following data was collected when studying the enzyme's activity at PH 6 and 25°C. The enzyme-catalyzed rate law follows the Michaelis-Menten model. Obtain the kinetic rate law from the experimental data.

v_0 (mM/s)	2.26	5.48	13.40	24.70
$[S]$ (M)	0.05	0.10	0.25	0.50
v_0 (mM/s)	40.90	62.30	94.30	105.00
$[S]$ (M)	1.00	2.50	5.00	8.00

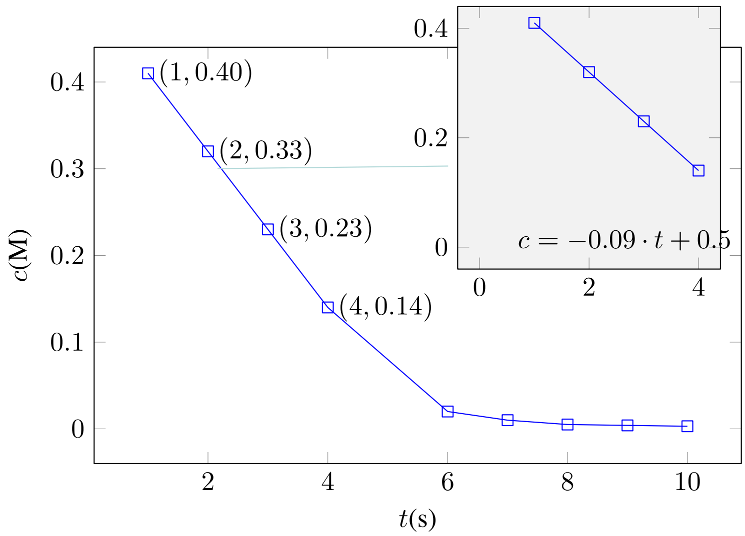
Solution

By inspecting the data, we can see that only the four first data points change linearly with time. After 4 seconds, the concentration reaches a plateau. In order to calculate initial velocities, we will use only the first four data points. We can calculate the average initial velocity by doing

$$\overline{r_0} = \frac{0.14 - 0.40}{4 - 1} = -0.086\text{M/s}$$

We can calculate the instantaneous initial velocity r_0 by interpolating the data with a line. You can do this, for example, using the www.graphpad.com/quickcalcs/linear1 web app. The equation of the line is: $c = -0.088 \cdot t + 0.5$ Hence, the instantaneous initial velocity

$$r_0 = \frac{\mathrm{d}c}{\mathrm{d}t} = -0.086\text{M/s}$$



Study Check

The following data was collected when studying an enzyme's activity at fixed PH and temperature. Obtain the Michaelis-Menten rate law from the experimental data.

v_0 (mM/s)	1.29	3.33	11.80	22.80	35.20	39.90	73.50	12.90
$[S]$ (M)	0.05	0.10	0.25	0.50	1.00	2.50	5.00	8.00

`PH and temperature effects` Enzyme activity depends on factors such as PH and temperature. PH has a strong impact on enzyme activity as activity passes through a maximum as PH increases. Similarly, enzyme activity also passes through a maximum as temperature increases. Enzymes show maximum activity at an *optimum PH* and *optimum temperature*. This maximum depends on the nature of the enzyme and the catalyzed reaction. Enzymes can lose activity reversively as well as irreversible. They reversibly lose activity at PH and temperature slightly different than their optimum. Enzyme deactivation is irreversible if the acidity or basicity is too high or if the temperature is too high or low. One explanation for reversible PH effects is that ionization of protein carboxylic and amino groups occurs when PH is not taken too far within the optimum PH. One explanation for irreversible PH effects is that enzymes lose their ternary structure hence losing their activity at very low or high PH values or temperatures higher than 30°C. Because of their shape, graphs displaying catalyst activity vs PH or temperature are often referred to as volcano plots. Figure **??** displays PH and temperature volcano plots for the activity of amylase.

CHAPTER 3

Rate of reaction

3.1 A is the reactant of two different reactions. Plot the data below in order to compare the rate of the two reaction.

	Reaction I	Reaction II
Time, s	$[A]$ (M)	$[A]$ (M)
0	100	100
1	7×10^{-1}	36.78
2	4×10^{-3}	13.53
3	3×10^{-5}	4.90
4	2×10^{-7}	1.80

3.2 Use the data below to identify reactants and products in a reaction.

Time, s	$[A]$ (M)	$[B]$ (M)
0	0	3
1	2×10^{-2}	2.4
2	3×10^{-2}	2.0
3	6×10^{-2}	1.64
4	8×10^{-2}	1.34

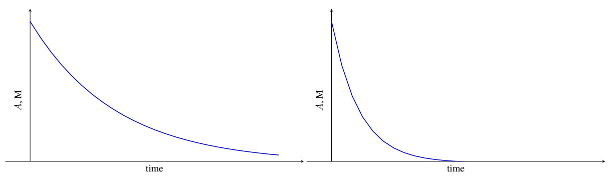
3.3 Use the data below to identify reactants and products in a reaction.

Time, s	$[A]$ (M)	$[B]$ (M)
0	2	0
1	1.996	4×10^{-3}
2	1.992	8×10^{-3}
3	1.988	1.2×10^{-2}
4	1.984	1.6×10^{-2}

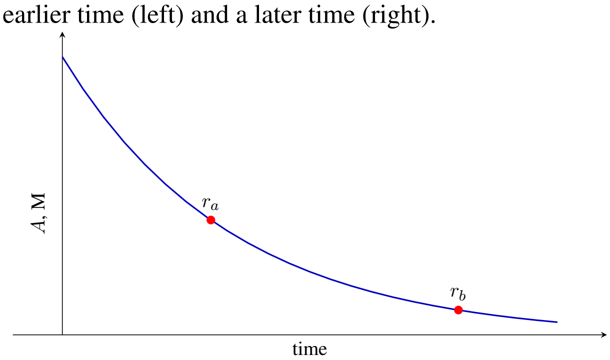
3.4 Use the data below to identify reactants and products in a reaction.

Time, s	$[A]$ (M)	$[B]$ (M)
0	4	0
1	3.6	2×10^{-3}
2	2.2	1×10^{-2}
3	1.3	1×10^{-2}
4	0.8	2×10^{-3}

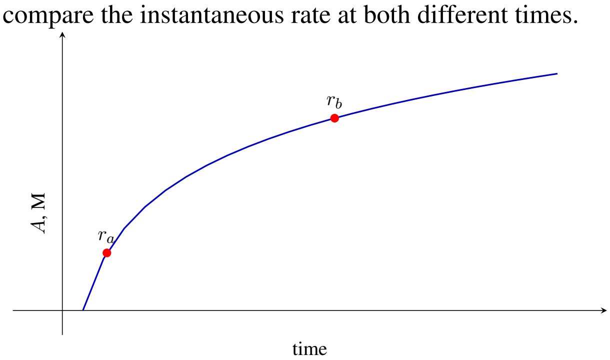
3.5 The concentration plots below represent two different reaction. Indicate which has the fastest initial reaction rate.



3.6 Using the concentration vs. time plot shown below, compare the instantaneous rate at both different times, an earlier time (left) and a later time (right).



3.7 Using the concentration vs. time plot shown below, compare the instantaneous rate at both different times.



3.8 For a given reaction the differential rate is measured for a set of species. Given the numerical values indicated next, identify the specie as reactant or product: (a) r_A=0.3M/s (b) r_B=-0.1M/s (c) r_C=-1.2M/s (d) r_D=0.6M/s

3.9 For the reaction below: $2\,a + 3\,b \longrightarrow 4\,c + 5\,d$ and given that the rate of disappearance of b is r_B=-0.3M/s, calculate: (a) the rate of disappearance of A, r_A (b) the rate of appearance of C, r_C (c) the rate of appearance of D, r_D (d) the rate of reaction r

3.10 For the reaction below: $2\,a + 2\,b \longrightarrow 3\,c + d$ and given that the rate of appearance of d is r_D=0.2M/s, calculate: (a) the rate of disappearance of A, r_A (b) the rate of disappearance of B, r_B (c) the rate of appearance of C, r_C (d) the rate of reaction r

Experimental measurement of reaction rates

3.11 Answer the following questions from the experimental data given in the table below (a) Plot the data. Which data points are suitable for calculating initial velocities? (b) Calculate the average initial velocity with four significant figures (c) Calculate the instantaneous initial velocity with four significant figures, indicating the r^2 correlation coeficient value

t (s)	1	2	3	4	6
c (M)	0.1750	0.2430	0.3150	0.3890	0.5000
t (s)	7	9			
c (M)	0.5100	0.5200			

3.12 Answer the following questions from the experimental data given in the table below (a) Plot the data. Which data points are suitable for calculating initial velocities? (b) Calculate the average initial velocity (c) Calculate the instantaneous initial velocity, indicating the r^2 correlation coeficient value

t (s)	1	2	3	4	6	7	9
c (M)	0.46	0.44	0.41	0.39	0.35	0.34	0.33

Rate laws

3.13 Given the following rate law

$$r\text{=0.04[A]}^2\text{[B][C]}^3$$

indicate: (a) the reaction order of A (b) the reaction order of B (c) the reaction order of C (d) the overall reaction order(e) the value of the reaction constant including its units

3.14 Given the following rate law

$$r\text{=0.4[Cl}_2\text{][F}_2\text{]}^2$$

indicate: (a) the reaction order of Cl_2 (b) the reaction order of F_2 (c) the overall reaction order(d) the value of the reaction constant including its units

3.15 Given the following rate law

$$r\text{=k[A]}^4\text{[B]}^2\text{[C]}^2$$

indicate the impact on the rate of the following: (a) to double [A] (b) to triple [B] (c) to double [C]

3.16 Given the following rate law

$$r\text{=k[A][B]}^2\text{[C]}$$

indicate the impact on the rate of the following: (a) to double [B] (b) to triple [A] (c) to quadruple [C]

3.17 Given the following rate constant values, identify the reaction order: (a) k=3 $\times$ 10^{-4} L/(s·mol) (b) k=0.03 M/s (c) k=3 $\times$ 10^{-5} 1/(s·M^2)

3.18 Given the following rate constant values, identify the reaction order: (a) k=0.04 M/s (b) k=0.00 1/s (c) k=0.09 1/s·M

3.19 Identify the following rate laws as integral or differential, and given a rate law form obtain the other corresponding type of form (i.e. if the integral from is given then obtain the differential form and viceversa): (a) $r = 0.0023$, $[A]_0$=0.3 (b) $Ln([\text{A}]) = -0.045 \cdot t + 0.45$ (c) $r = 0.3[A]^2$, $[A]_0$=25

3.20 Identify the following rate laws as integral or differential, and given a rate law form obtain the other corresponding type of form (i.e. if the integral from is given then obtain the differential form and viceversa):

(a) [A] $= 0.045 - 0.34 \cdot t$ (b) $r = 0.9[A]^1$, $[A]_0$=1.49 (c) $\frac{1}{[A]} = 0.3 + 0.04 \cdot t$

3.21 Compute the half-life for the following reactions: (a) a first order reaction with k=0.234 1/s (b) a zeroth-order reaction with k=0.34 M/s and initial concentration 0.1M (c) a second-order reaction with k=0.067 M/s and initial concentration 0.01M

3.22 Compute the half-life for the following reactions: (a) a zeroth-order reaction with k=0.045 M/s and initial concentration 0.05M (b) a first order reaction with k=0.947 1/s and initial concentration 0.05M (c) a second-order reaction with k=0.0087 M/s and initial concentration 0.087M

THE DIFFERENTIAL AND INTEGRAL METHODS OF OBTAINING RATE LAWS

3.23 Use the data below to calculate the rate law of the following reaction:

$$\text{A} \longrightarrow \text{B}$$

Experiment	r (M/s)	$[A]$, (M)
1	0.03	1
2	0.12	2

3.24 Use the data below to calculate the rate law of the following reaction:

$$\text{A} \longrightarrow \text{B}$$

Experiment	r (M/s)	$[A]$, (M)
1	5×10^{-3}	0.5
2	5×10^{-3}	0.6

3.25 Use the data below to calculate the rate law of the following reaction:

$$\text{A} \longrightarrow \text{B}$$

Experiment	r (M/s)	$[A]$, (M)
1	0.03	0.01
2	0.30	0.1

3.26 Use the data below to calculate the rate law of the following reaction:

$$\text{A} \longrightarrow \text{B}$$

Experiment	r (M/s)	$[A]$, (M)
1	0.5	0.253
2	1	0.320

3.27 Use the data below to calculate the rate law:

Experiment	$[A]$ (M)	$[B]$ (M)	r (M/s)
1	0.1	0.02	2.08×10^{-5}
2	0.1	0.04	8.32×10^{-5}
3	0.2	0.01	1.04×10^{-5}
4	0.4	0.01	2.08×10^{-5}

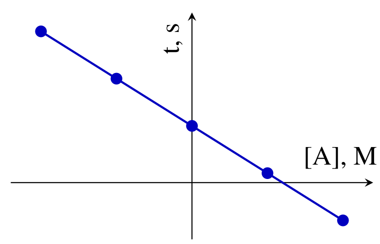
3.28 Use the data below to calculate the rate law:

Experiment	$[A]$ (M)	$[B]$ (M)	r (M/s)
1	0.1	0.02	1.03×10^{-3}
2	0.1	0.04	2.08×10^{-3}
3	0.2	0.01	1.04×10^{-3}
4	0.4	0.01	2.08×10^{-3}

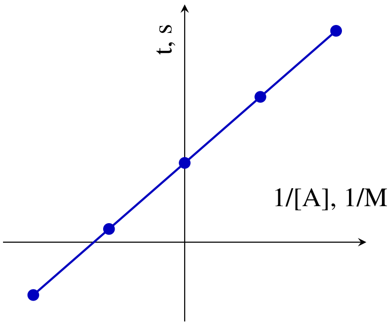
3.29 Use the data below to calculate the rate law:

Experiment	$[A]$ (M)	$[B]$ (M)	r (M/s)
1	0.1	0.02	1.03×10^{-3}
2	0.1	0.04	2.08×10^{-3}
3	0.2	0.01	1.04×10^{-3}
4	0.4	0.01	2.08×10^{-3}

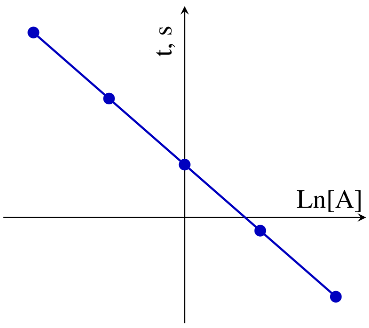
3.30 The plot below resulting from integral method represent a perfect line with r^2=0.99. Indicate the order of the reaction.



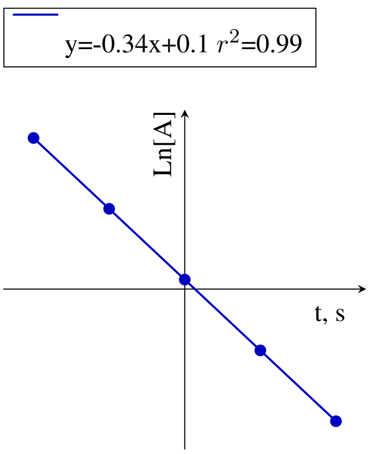
3.31 The plot below resulting from integral method represent a perfect line with r^2=0.99. Indicate the order of the reaction.



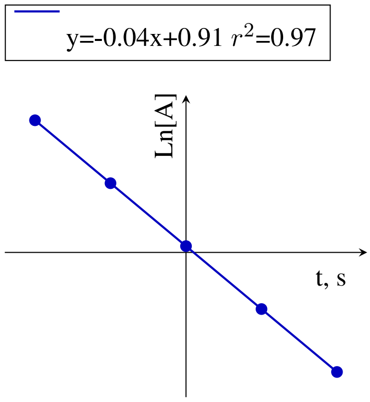
3.32 The plot below resulting from integral method represent a perfect line with r^2=0.99. Indicate the order of the reaction.



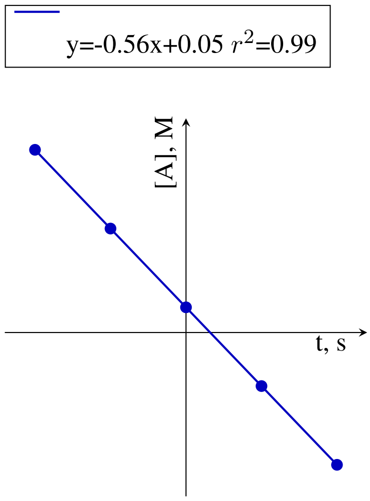
3.33 The following plot results from processing data by means of the integral method. Interpret the linear regressions and indicate the rate law.



3.34 The following plot results from processing data using the integral method. Interpret the linear regressions, paying particular attention to the statistics behind the regression, and indicate the rate law.



3.35 The following plot results from processing data by means of the integral method. Interpret the linear regressions and indicate the rate law.



3.36 Using the following data, calculate the order and rate constant and write down the rate law.

t (s)	$[A]$ (M)
5	0.952
10	0.625
15	0.465
20	0.370
25	0.308
35	0.230

3.37 Using the following data, calculate the order and rate constant and write down the rate law.

t (s)	$[A]$ (M)
0	9.90×10^{-2}
5	4.97×10^{-2}
10	3.32×10^{-2}
15	2.49×10^{-2}
20	1.66×10^{-2}
25	1.43×10^{-2}

3.38 Calculate the rate law for the following reaction using the given data: $2\,ClO_2 + 2\,I^- \longrightarrow 2\,ClO_2{}^- + I_2$

t (s)	$[ClO_2]$ (M)
0	4.77×10^{-4}
1	3.31×10^{-4}
2	3.91×10^{-4}
3	3.53×10^{-4}
5	2.89×10^{-4}
10	1.76×10^{-4}
30	2.40×10^{-5}
50	3.20×10^{-6}

3.39 Calculate the rate law for the following reaction using the given data: $CH_3COOCH_3 + OH^- \longrightarrow CH_3COO^- + CH_3OH$

t (s)	$[CH_3COOCH_3]$ (M)
0	1.00×10^{-2}
3	7.40×10^{-3}
4	6.83×10^{-3}
5	6.34×10^{-3}
10	4.63×10^{-3}
20	3.04×10^{-3}
30	2.24×10^{-3}

3.40 Calculate the rate law for the following reaction using the given data: $2\,NO_2 \longrightarrow 2\,NO + O_2$

t (s)	$[NO_2]$ (M)
0	1.00×10^{-2}
60	6.83×10^{-3}
120	5.18×10^{-3}
180	4.18×10^{-3}
240	3.50×10^{-3}
300	3.01×10^{-3}
360	2.64×10^{-3}

3.41 Calculate the rate law for the following reaction using the given data: $2\,HI \longrightarrow H_2 + I_2$

t (s)	[HI] (M)
0	1
1000	0.11
2000	0.061
3000	0.041
4000	0.031

3.42 Calculate the rate law for the following reaction using the given data: $H_2 + I_2 \longrightarrow 2\,HI$

t (s)	$[H_2]$, (M)
0	1
1	0.43
2	0.27
3	0.20
4	0.16

3.43 Calculate the rate law for the following reaction using the given data:$C_3H_5 \longrightarrow C_3H_6$

t (s)	$[C_3H_6]$ (M)
0	1.48×10^{-3}
5	1.25×10^{-3}
10	1.03×10^{-3}
15	8.05×10^{-4}
20	5.80×10^{-4}

3.44 Calculate the rate law for the following reaction using the given data: $2\,P + Q \longrightarrow W$

t (s)	[P] (mM)
0	1
15	0.97484
30	0.9504
45	0.92676
60	0.90379
75	0.881
90	0.8596

3.45 Calculate the rate law for the following reaction using the given data: $2\,O_3 \longrightarrow 3\,O_2$

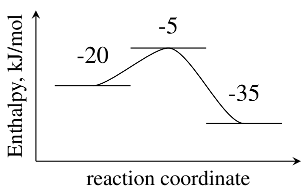
t (h)	$[O_3]$ (M)
5	0.952
10	0.625
15	0.465
20	0.37
25	0.308
35	0.23

3.46 Calculate the rate law for the following reaction using the given data: $2\,X \longrightarrow Y + Z$

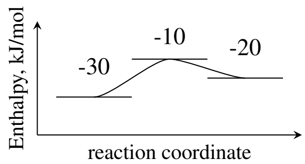
t (s)	[X] (M)
5	0.0990
10	0.0497
15	0.0332
20	0.0249
25	0.0200
30	0.0166
35	0.0143
40	0.0125

Collision theory

3.47 For the energy profile below, indicate the activation energy and the reaction energy. Does this diagram corresponds to an exothermic or endothermic reaction?



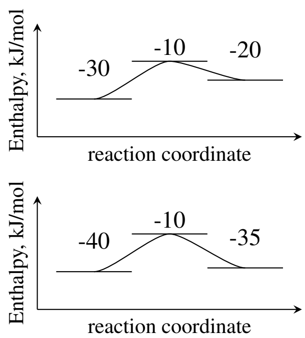
3.48 For the energy profile below, indicate the activation energy and the reaction energy. Does this diagram corresponds to an exothermic or endothermic reaction?



3.49 Sketch an energy profile in which reactants are located at -10kJ/mol, the activation energy is 15kJ/mol and the energy of reaction is 10kJ/mol.

3.50 Sketch an energy profile in which reactants are located at -15kJ/mol, the activation energy is 10kJ/mol and the energy of reaction is -10kJ/mol.

3.51 Compare both energy profiles top and bottom and indicate: (a) What reaction proceeds faster (b) What reaction exchanges more energy



3.52 The reaction energy of a process, going from reactants into products, is 10kJ/mol and its activation energy is 15kJ/mol. Calculate the reaction energy and the activation energy of the reverse process, starting from products and going to reactants. Is this reverse reaction exothermic or endothermic. How about the direct reaction, starting from reactants and going to products?

3.53 The rate constant for the gas-phase decomposition of N_2O has the following temperature dependence:

$$2\,N_2O_{(g)} \longrightarrow 2\,N_{2(g)} + O_{2(g)}$$

has the following temperature dependence:

T (K)	k, (s^{-1})
1500	9214
1400	2175
1300	411
1200	58.9

Compute the activation energy in (kJ/mol) and the frequency factor. Can you guess the reaction order?

3.54 For the gas-phase reaction between hydrogen and iodine

$$H_{2(g)} + I_{2(g)} \longrightarrow 2\,HI_{(g)}$$

the rate constant at 500K was found to be 9.36×10^{-7} 1/(M·s) and the rate constant at 600K was found to be 5.69×10^{-4} 1/(M·s). Calculate the activation energy in kJ/mol and the frequency factor of the reaction. Can you guess the reaction order?

3.55 For the decomposition of dinitrogen pentoxide

$$2\,N_2O_{5(g)} \longrightarrow 2\,N_2O_{4(g)} + O_{2(g)}$$

the rate constant at 500K is 345.43 s^{-1} and at different temperature, the rate constant is 24.84 s^{-1}. Given that the activation energy of this reaction is 103.03 kJ/mol, calculate the second temperature value.

3.56 For a given reaction, when temperature increases from 100K to 200K, its rate constant triples. Calculate the activation energy for this process.

Reaction mechanisms

3.57 For the following elemental reaction steps indicate the molecularity and give the rate law: (a) $N_2O \longrightarrow NO + O$ (b) $N_2O + O \longrightarrow N_2 + O_2$(c) $2\,NO \longrightarrow N_2O_2$(d) $N_2O_2 + H \longrightarrow N_2O + HO$

3.58 For the following elemental reaction steps indicate the molecularity and give the rate law: (a) $SiH_4 + SiH_2 \longrightarrow Si_2H_6$ (b) $SiH_2 + H \longrightarrow SiH + H_2$ (c) $Si_2H_3 + H_2 \longrightarrow Si_2H_5$

3.59 The experimental rate law for the reaction of nitrogen monoxide and molecular oxygen given by the following reaction

$$2\,NO_{(g)} + O_{2(g)} \longrightarrow 2\,NO_{2(g)} \qquad r^{Exp}{=}k^{Exp}\,[NO]^2[O_2]$$

is second order with respect to dinitrogen oxide and first order with respect to oxygen. A plausible mechanism would be:

(1) $$NO_{(g)} + NO_{(g)} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} N_2O_{2(g)}$$

(2) $$N_2O_{2(g)} + O_{2(g)} \xrightarrow{k_2} 2\,NO_{2(g)} \qquad \text{(slow)}$$

Calculate: (a) the molecularity of each reaction step (b) the rate law for each reaction step (c) identify all intermediates (d) justify whether the suggested mechanism is valid and if so give the expression for the apparent reaction constant k^{Exp} in terms of the rate constants in the mechanism (e) give the expression for the apparent activation energy in terms of the activation energies of each step involved in the apparent constant

3.60 A possible mechanism for the reaction of nitrogen monoxide and molecular oxygen to produce nitrogen dioxide given by the following reaction

$$2\,NO_{(g)} + O_{2(g)} \longrightarrow 2\,NO_{2(g)}$$

could be:

(1) $$NO_{(g)} + O_{2(g)} \xrightarrow{k_1} NO_{3(g)} \qquad \text{(slow)}$$

(2) $$NO_{3(g)} + NO_{(g)} \xrightarrow{k_2} 2\,NO_{2(g)} \qquad \text{(fast)}$$

Calculate: (a) the molecularity of each reaction step (b) the rate law for each reaction step (c) identify all intermediates (d) give the experimental rate law

3.61 For a mechanism, the expression for the apparent reaction cound was found to be

$$r^{Exp} = k_1 \cdot k_2 \cdot k_{-1}$$

given that the numerical values of the activation energies for the steps one, one in reverse and two are respectively 100, 90 and 200 in $kJ \cdot mol^{-1}$, calculate: (a) the value of the apparent activation energy in $kJ \cdot mol^{-1}$ (b) the value of the apparent activation energy in eV

3.62 For a mechanism, the expression for the apparent reaction cound was found to be

$$r^{Exp} = \frac{k_1 \cdot k_{-1}}{k_2}$$

given that the numerical values of the activation energies for the steps one, one in reverse and two are respectively 100, 90 and 200 in $kJ \cdot mol^{-1}$, calculate: (a) can the activation energy for a given step in the mechanism be negative? (b) the value of the apparent activation energy in $kJ \cdot mol^{-1}$. Is this value reasonable? (c) the value of the apparent activation energy in eV

Catalysis by enzymes

3.63 The measured initial velocity for an enzyme-catalyzed reaction that follows the Michaelis-Menten model was 3×10^{-3} mM/min, whereas the maximum velocity was 5×10^{-3} mM/min. Calculate Michaelis-Menten's k_m parameter for a substrate concentration of 3×10^{-2}mM.

3.64 A solution of urease, an enzyme that catalyzes the hydrolysis of urea into ammonia and carbon dioxide, is mixed with a 0.01mM urea solution. The measured initial velocity for the urea decomposition was 1×10^{-3} mM/min. Calculate the maximum velocity of the enzymatic reaction given that under the experimental conditions, K_m was found to be 0.05mM.

3.65 For an enzyme obeying Michaelis-Menten kinetic model, at what substrate concentration will the initial velocity be equal to half of r_m?

3.66 For an enzyme obeying Michaelis-Menten kinetic model, at what substrate concentration will the initial velocity be equal 25% of r_m?

3.67 For an enzyme-catalyzed reaction obeying Michaelis-Menten kinetics, r_m was 1.0×10^{-9} M/s and K_m = 2.0×10^{-3} M. Calculate the rate of the enzymatic reaction for a substrate concentration of 1.0×10^{-3} M.

3.68 For an enzyme-catalyzed reaction obeying Michaelis-Menten kinetics, r_m was 1.0×10^{-8} M/s and K_m = 5.0×10^{-3} M. Calculate the rate of this enzymatic reaction when the substrate concentration was 1.0×10^{-2} M.

3.69 Lysozyme is an enzyme that protects against bacterial, fungal, and viral pathogens. The following kinetic data reflects the activity of lysozyme acting on the hexa-N-acetylglucosamine substrate at controlled PH and temperature. Adjust the Lineweaver-Burk rate law to the data and obtain the two Michaelis-Menten constants.

$r_0/10^{-4}$ (μM/s)	8.2	16	24	31	38
$[S]$ (μM)	0.1	0.2	0.3	0.4	0.5

3.70 Pyruvate carboxylase is a key enzyme in metabolism. The following kinetic data reflects the enzyme's activity at controlled PH and temperature. Adjust the Lineweaver-Burk rate law to the data and obtain the two Michaelis-Menten constants.

$r_0/10^{-5}$ (μM/s)	1.6	3.3	4.9	6.6	8.2
$[S]$ (μM)	0.1	0.2	0.3	0.4	0.5

Answers **3.1** Reaction I is faster **3.2** B is the reactants and A the product **3.3** A is the reactants and B the product **3.4** A is the reactants and B an intermediate, neither reactant or product **3.5** The plot on the right **3.6** $r_{left} > r_{right}$ **3.7** $r_{left} > r_{right}$ **3.8** (a) r_A=0.3M/s (product) (b) r_B=-0.1M/s (reactant) (c) r_C=-1.2M/s (reactant) (d) r_D=0.6M/s (product) **3.9** (a) - 0.2M/s (b) +0.4M/s (c) +0.5M/s (d) 0.1M/s **3.10** (a) - 0.4M/s (b) - 0.4M/s (c) +0.6M/s (d) 0.2M/s **3.11** (a) first four points (b) 0.0714M/s (c) 0.0713M/s; 0.99 **3.12** (a) first four points (b) -0.024M/s (c) -0.023M/s; 0.99 **3.13** (a) second order (b) first order (c) third order (d) sixth order (e) 0.04 s/M^5 **3.14** (a) first order (b) second order (c) third order (d) 0.4 s/M^2 **3.15** (a) 16 times faster (b) 9 times faster (c) 4 times faster **3.16** (a) 4 times faster (b) 3 times faster (c) 4 times faster **3.17** (a) second order (b) zeroth order (c) third order **3.18** (a) zeroth order (b) first order (c) second order **3.19** (a) differential, $[\mathrm{A}] = -0.0023 \cdot t + 0.3$ (b) integral, $r = 0.045[A]^1$, $[A]_0$=1.56 (c) differential, $\frac{1}{[\mathrm{A}]} = 0.3 \cdot t + 0.04$ **3.20** (a) integral, $r = 0.34$, $[A]_0$=0.045 (b) differential, $Ln([\mathrm{A}]) = 0.4 - 0.9 \cdot t$ (c) integral, $r = 0.04[A]^2$, $[A]_0$=3.33 **3.21** (a) 2.96s (b) 0.15 s (c) 1492 s **3.22** (a) 0.55 s (b) 0.73s (c) 1321.1 s **3.23** $r = 0.03[A]^2$ **3.24** $r = 0.005$ **3.25** $r = 3[A]$ **3.26** $r = 30.5[A]^3$ **3.27** $r = 0.52[A]^1[B]^2$ **3.28** $r = 0.52[A][B]$ **3.29** $r = 0.52[A][B]$ **3.30** zeroth-order **3.31** second order **3.32** first order **3.33** $r = 0.34[A]^1$, $[\mathrm{A}]_0$=1.1M **3.34** No rate law can be deribed **3.35** $r = 0.56$, $[\mathrm{A}]_0$=0.05M **3.36** $r = 0.11[A]^2$, $[\mathrm{A}]_0$=0.5M **3.37** $r = 2.45[A]^2$, $[\mathrm{A}]_0$=0.13M **3.38** $r = 0.098[\mathrm{ClO_2}]$, $[\mathrm{ClO_2}]_0$=4.5×10^{-4}M **3.39** $r = 11.51[\mathrm{CH_3COOCH_3}]^2$, $[\mathrm{CH_3COOCH_3}]_0$=9.9×10^{-3}M **3.40** $r = 0.77[\mathrm{NO_2}]^2$, $[\mathrm{NO_2}]_0$=0.01M **3.41** $r = 7.69 \times 10^{-3}[\mathrm{HI}]^2$, $[\mathrm{HI}]_0$=0.7M **3.42** $r = 1.31[\mathrm{H_2}]^2$, $[\mathrm{H_2}]_0$=1.02M **3.43** $r = 4.5 \times 10^{-5}$, $[\mathrm{C_3H_6}]_0$=1.48×10^{-3}M **3.44** $r = 640.33$, $[\mathrm{P}]_0$=639.2mM **3.45** $r = 0.109[\mathrm{O_3}]^2$, $[\mathrm{O_3}]_0$=1.99M **3.46** $r = 1.99[\mathrm{X}]^2$, $[\mathrm{X}]_0$=6.15M **3.47** E_a =15kJ/mol; ΔH =-15kJ/mol; exothermic **3.48** E_a =20kJ/mol; ΔH =10kJ/mol; endothermic **3.49** Reactants (-10); TS (5); products(0) **3.50** Reactants (-15); TS (-5); products(-25) **3.51** (a) top goes faster (b) top consumes more energy **3.52** E_a =5kJ/mol; ΔH =-10kJ/mol; direct endothermic; reverse exothermic **3.53** E_a =252.05kJ/mol; $A = 5.52 \times 10^{12}$ s^{-1}; first based on units of k. **3.54** E_a =159.88kJ/mol; $A = 4.73 \times 10^{10}$ s^{-1}. Second order based on units of k. **3.55** 452K **3.56** 1.8kJ/mol **3.57** (a) unimolecularity; r=k[$\mathrm{N_2O}$] (b) bimolecular; r=k[$\mathrm{N_2O}$][O] (c) bimolecular; r=k[NO]^2 (d) bimolecular; r=k[$\mathrm{N_2O_2}$][H] **3.58** (a) bimolecular; r=k[$\mathrm{SiH_2}$][$\mathrm{SiH_4}$] (b) bimolecular; r=k[$\mathrm{SiH_2}$][H] (c) bimolecular; r=k[$\mathrm{Si_2H_3}$][$\mathrm{H_2}$] **3.59** (a) (2;2) (b) $0 = k_1[\mathrm{NO}]^2 - k_{-1}[\mathrm{N_2O_2}]$; $r_2 = k_2[\mathrm{N_2O_2}][\mathrm{O_2}]$ (c) $\mathrm{N_2O_2}$ (d) it is valid; $\frac{k_2 \cdot k_1}{k_{-1}}$ (e) $E_{a_2} + E_{a_1} - E_{a_{-1}}$ **3.60** (a) bimolecular; bimolecular (b) $r_1 = k_1[\mathrm{NO}][\mathrm{O_2}]$; $r_2 = k_2[\mathrm{NO_3}][\mathrm{NO}]$ (c) $\mathrm{NO_3}$ (d) $r^{Exp} \sim r^{slow} = k_1[\mathrm{NO}][\mathrm{O_2}]$ **3.61** (a) 390$kJ \cdot mol^{-1}$ (b) 4eV **3.62** (a) no (b) -10$kJ \cdot mol^{-1}$; yes an apparent activation energy can be negative (c) 0.1eV **3.63** 0.02mM **3.64** 1×10^{-3} mM/min **3.65** $[S] = K_m$ **3.66** $[S] = 1/3K_m$ **3.67** 3.3×10^{-10}M/s **3.68** 6.6×10^{-9}M/s **3.69** $r_m = 0.05\mu$ M/s;$K_m = 5.34\mu$M **3.70** $r_m = 0.01\mu$ M/s;$K_m = 60\mu$M